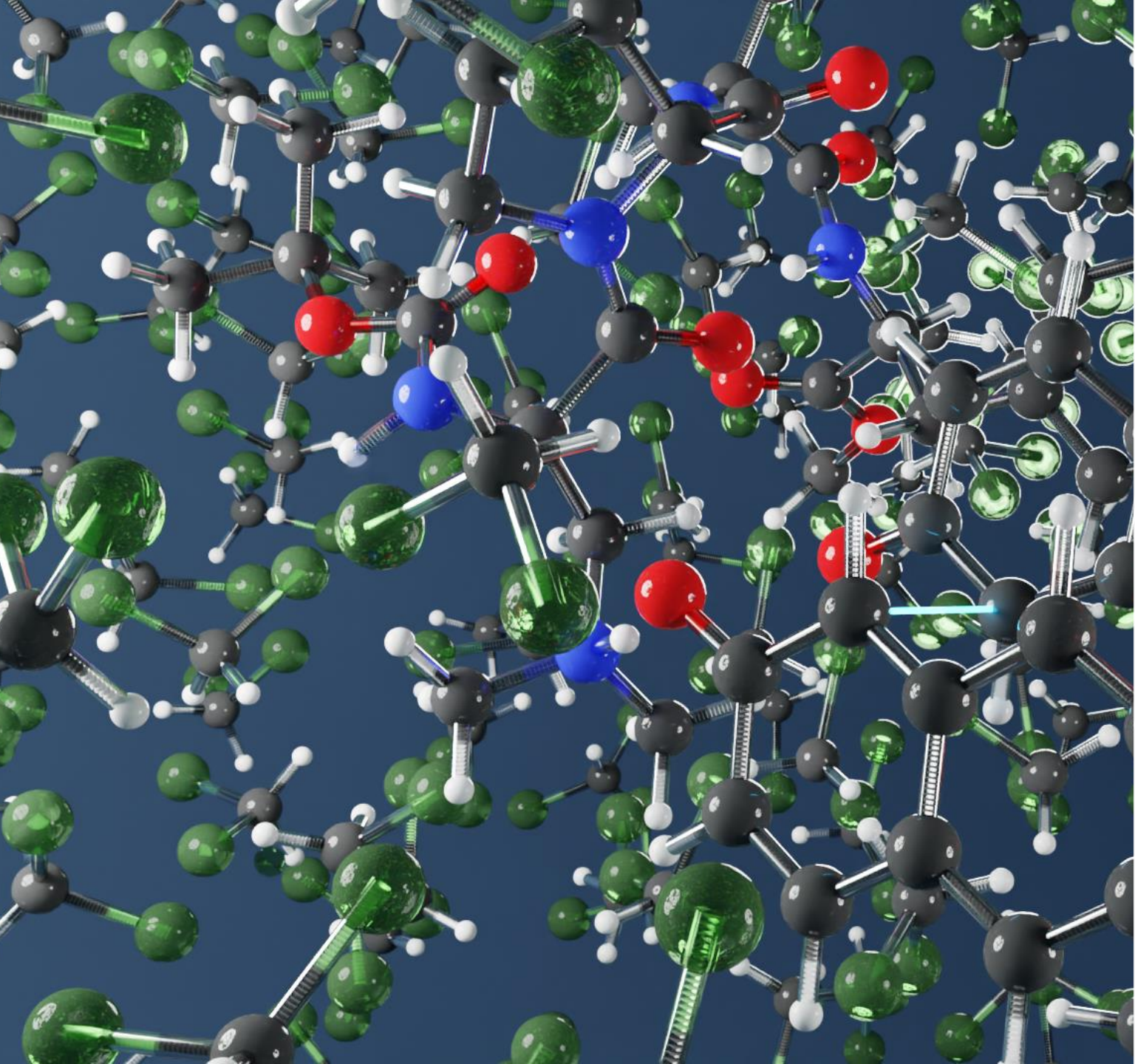
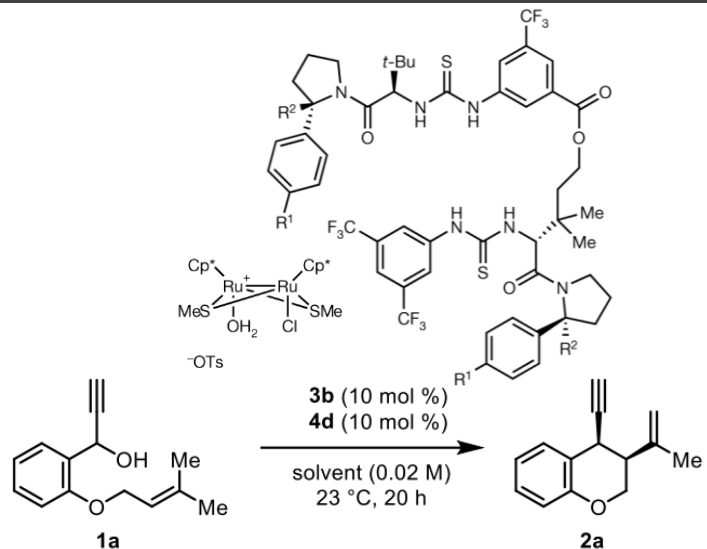




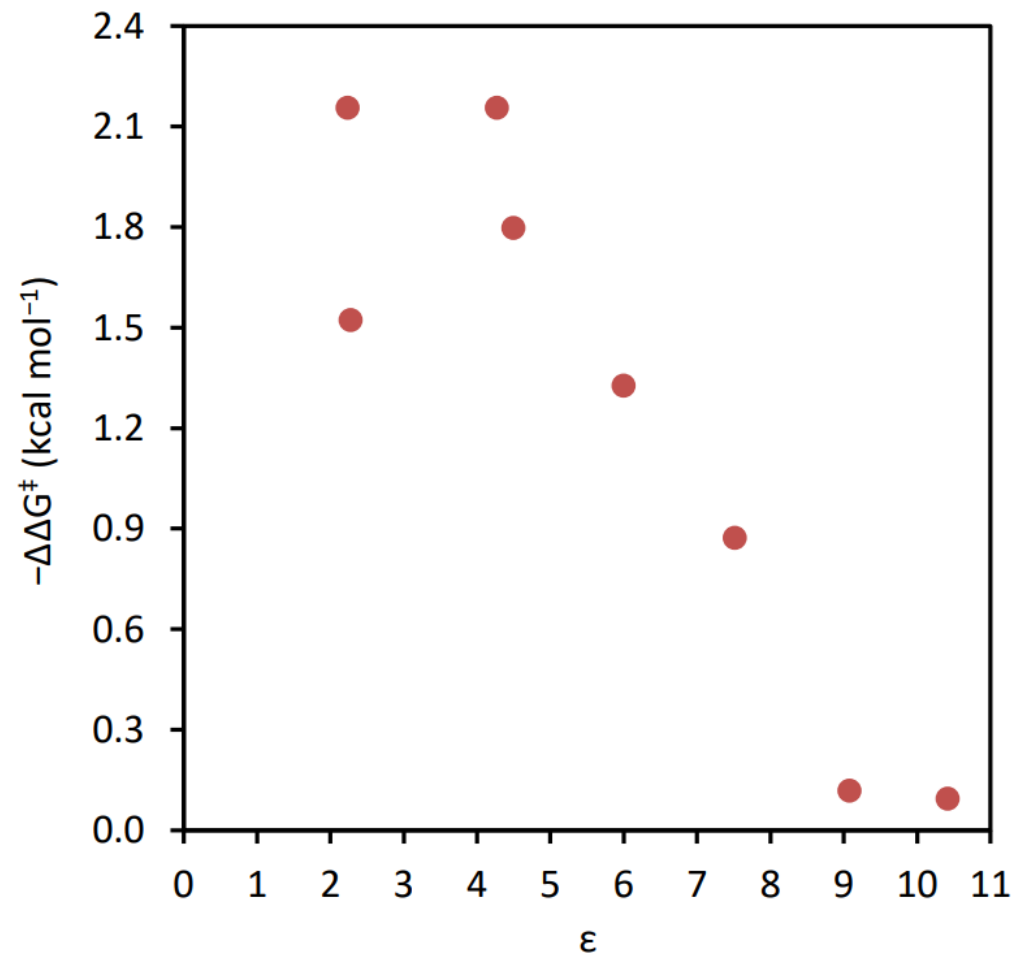
Solvent Effects on Chemical Reactivity and Catalysis

Nov 2024
Literature Meeting

Studies of solvent effects can provide insight, but are often auxiliary and qualitative



solvent	ϵ	dr	ee (%)	$-\Delta\Delta G^\ddagger$ (kcal mol ⁻¹)
CCl ₄	2.24	35:1	95	2.16
9:1 CCl ₄ :DCM	—	30:1	93	—
benzene	2.28	12:1	86	1.52
19:1 benzene:DCM	—	11:1	85	—
Et ₂ O	4.27	36:1	95	2.16
9:1 Et ₂ O:DCM	—	33:1	94	—
TBME	4.50	31:1	91	1.80
EtOAc	6	12:1	81	1.33
THF	7.52	6:1	63	0.87
DCM	9.08	5:1	10	0.12
1,2-DCE	10.42	4:1	8	0.09



Motivating questions

Why are the outcomes of some reactions highly but subtly solvent-dependent?

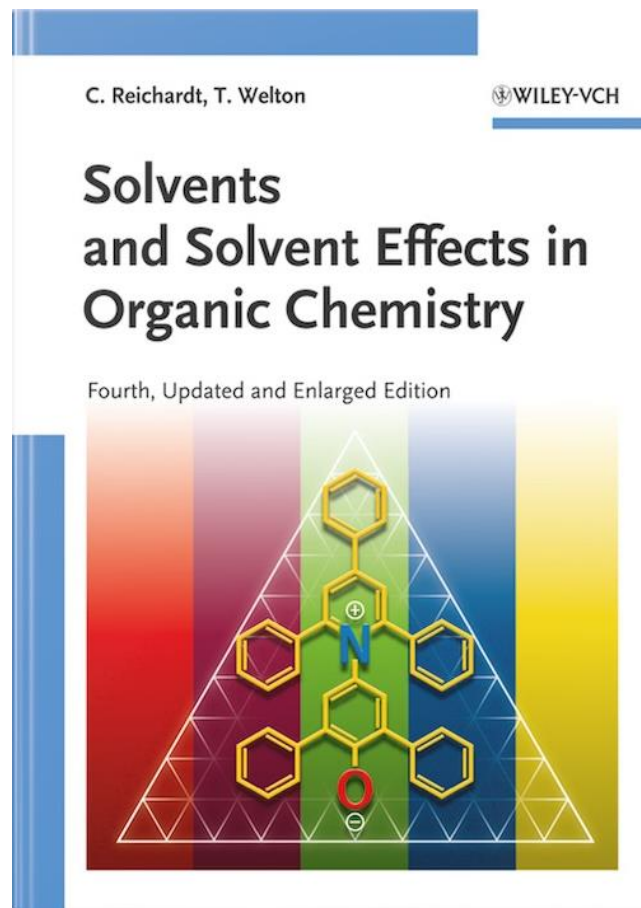
When are solvent effects a dominant driver of reactivity/selectivity?

Is the analogy between a catalyst and the first solvation shell useful?

Are solvent effects (in catalysis) amenable to rigorous investigation?

Can we leverage an understanding of solvent effects towards more efficient catalysts?

Fundamentals of solvent effects



Intermolecular interactions underlie solvent effects

1. Ion-dipole

$$U_{\text{ion-dipole}} = -\frac{1}{4\pi \cdot \epsilon_0} \cdot \frac{z \cdot e \cdot \mu \cdot \cos \theta}{r^2}$$

2. Dipole-dipole

$$U_{\text{dipole-dipole}} = -\frac{1}{(4\pi \cdot \epsilon_0)^2} \cdot \frac{2\mu_1^2 \cdot \mu_2^2}{3k_B \cdot T \cdot r^3}$$

3. Dipole-induced-dipole

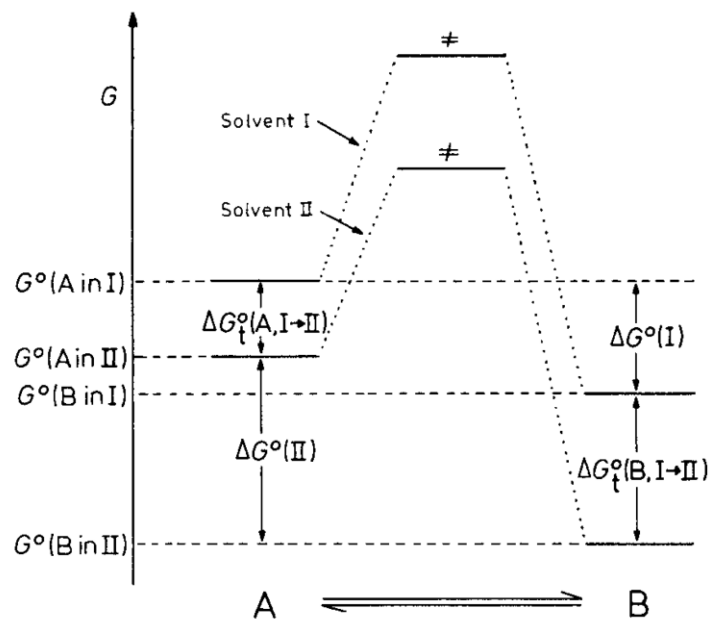
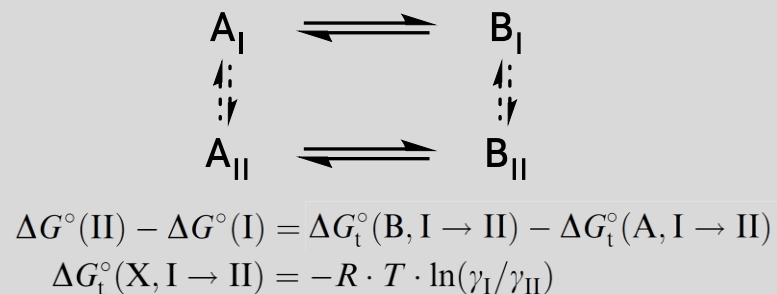
$$U_{\text{dipole-induced dipole}} = -\frac{1}{(4\pi \cdot \epsilon_0)^2} \cdot \frac{\alpha_1 \cdot \mu_2^2 + \alpha_2 \cdot \mu_1^2}{r^6}$$

4. London dispersion

$$U_{\text{dispersion}} = -\frac{1}{(4\pi \cdot \epsilon_0)^2} \cdot \frac{3\alpha_1 \cdot \alpha_2}{2r^6} \cdot \left(\frac{I_1 \cdot I_2}{I_1 + I_2} \right)$$

Solvents can shift the positions of thermodynamic equilibria

Consider the equilibrium between **A** and **B** in two solvents **I** and **II**:



If **B** is more soluble in solvent **II** (and A is equally soluble in both) then the equilibrium shifts towards **B** in solvent **II**.

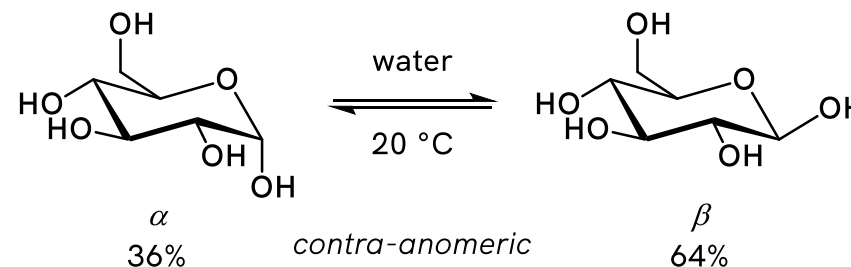
Common phenomena

1. Acidity/basicity

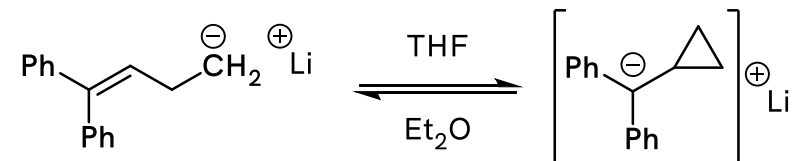
in aqueous solution: $t\text{-BuOH} < i\text{-PrOH} < \text{MeOH}$

in gas phase: $\text{MeOH} < i\text{-PrOH} < t\text{-BuOH}$

2. Conformation

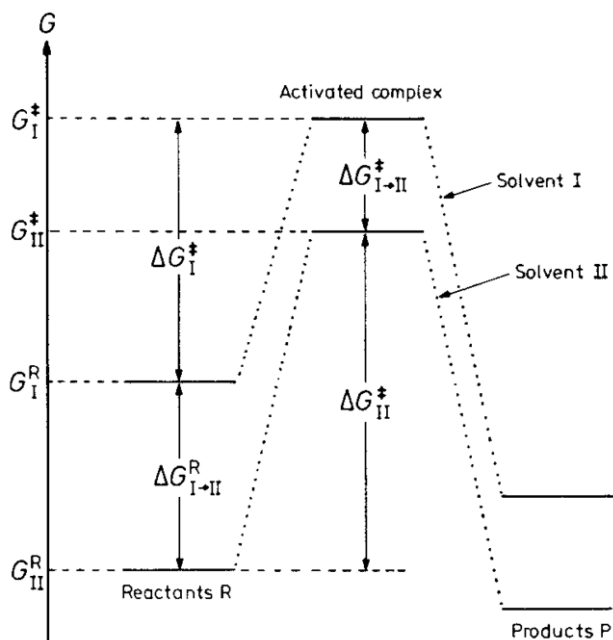
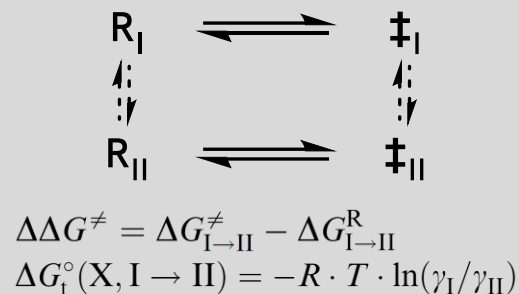


3. Valence isomerism/electron transfer



Solvents can change reaction rates through differential TS/GS solvation

Consider the equilibrium between **R** and **‡** in two solvents **I** and **II**:

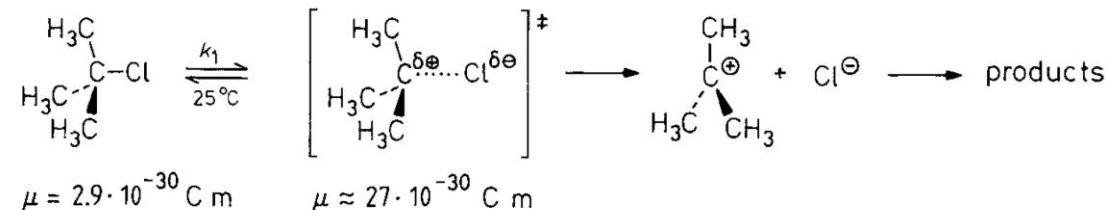


This requires that both **R** and **‡** are in equilibrium with the solvent.

Hughes-Ingold rules

For a reaction in which **‡** has a larger dipole than **R**, a more polar solvent effects a higher rate.

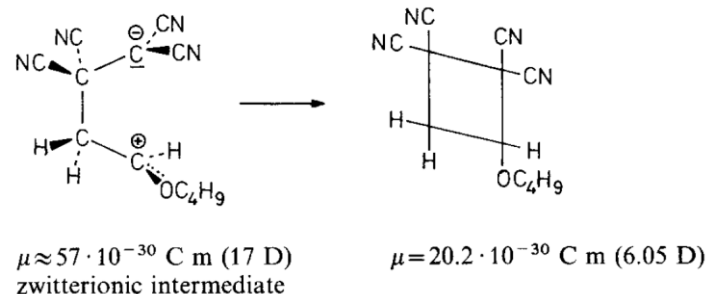
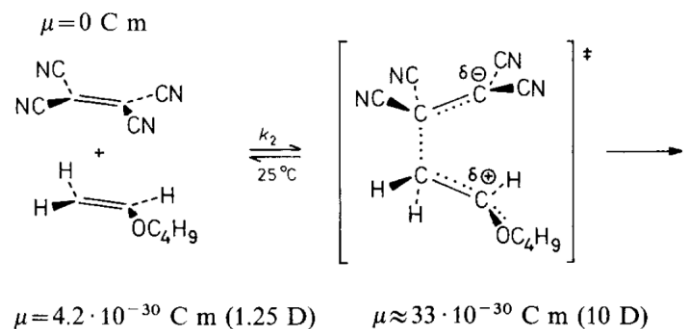
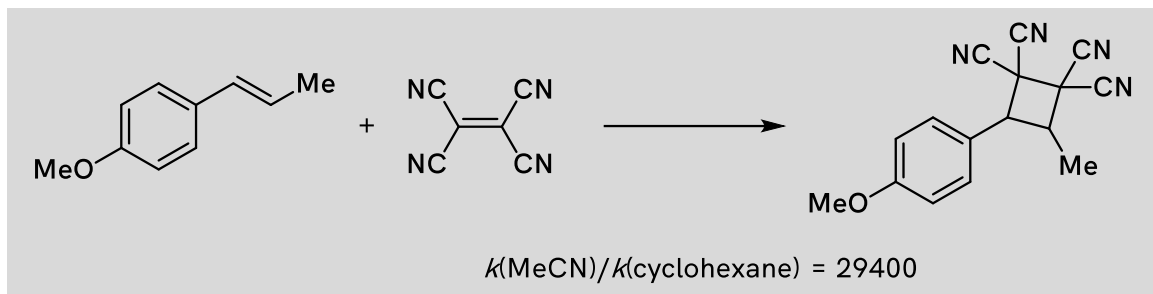
Reaction type	Initial reactants	Activated complex	Charge alteration during activation	Effect of increased solvent polarity on rate ^{a)}
(a) S _N 1	R—X	R ^{δ+} ... X ^{δ-}	Separation of unlike charges	Large increase
(b) S _N 1	R—X ⁺	R ^{δ+} ... X ^{δ+}	Dispersal of charge	Small decrease
(c) S _N 2	Y + R—X	Y ^{δ+} ... R ... X ^{δ-}	Separation of unlike charges	Large increase
(d) S _N 2	Y ⁻ + R—X	Y ^{δ-} ... R ... X ^{δ-}	Dispersal of charge	Small decrease
(e) S _N 2	Y + R—X ⁺	Y ^{δ+} ... R ... X ^{δ+}	Dispersal of charge	Small decrease
(f) S _N 2	Y ⁻ + R—X ⁺	Y ^{δ-} ... R ... X ^{δ+}	Destruction of charge	Large decrease



Solvent	C ₂ H ₅ OH	CH ₃ OH	HCONH ₂	HCO ₂ H	H ₂ O
$k_1^{\text{rel}*)}$	1	9	430	12200	335000

Hughes–Ingold rules are qualitatively general

Solvent effects as mechanistic probes:

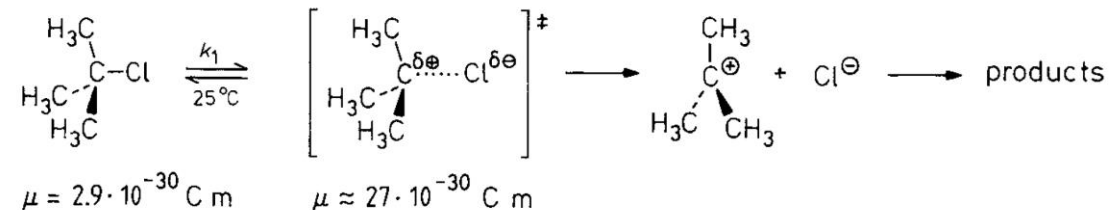


Hughes–Ingold fails when specific solvation effects are involved,
or when TST is not valid

Hughes–Ingold rules

For a reaction in which $\ddagger$ has a larger dipole than **R**, a more polar solvent effects a higher rate.

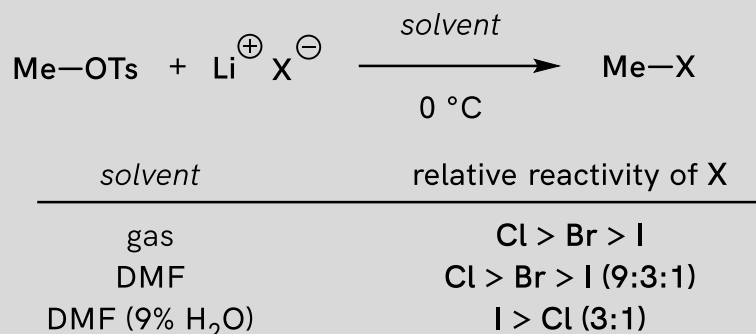
Reaction type	Initial reactants	Activated complex	Charge alteration during activation	Effect of increased solvent polarity on rate ^{a)}
(a) S_N1	$R-X$	$R^{\delta\oplus} \dots X^{\delta\ominus}$	Separation of unlike charges	Large increase
(b) S_N1	$R-X^{\oplus}$	$R^{\delta\oplus} \dots X^{\delta\oplus}$	Dispersal of charge	Small decrease
(c) S_N2	$Y + R-X$	$Y^{\delta\oplus} \dots R \dots X^{\delta\ominus}$	Separation of unlike charges	Large increase
(d) S_N2	$Y^{\ominus} + R-X$	$Y^{\delta\ominus} \dots R \dots X^{\delta\ominus}$	Dispersal of charge	Small decrease
(e) S_N2	$Y + R-X^{\oplus}$	$Y^{\delta\oplus} \dots R \dots X^{\delta\oplus}$	Dispersal of charge	Small decrease
(f) S_N2	$Y^{\ominus} + R-X^{\oplus}$	$Y^{\delta\ominus} \dots R \dots X^{\delta\oplus}$	Destruction of charge	Large decrease



Solvent	C_2H_5OH	CH_3OH	$HCONH_2$	HCO_2H	H_2O
$k_1^{rel*})$	1	9	430	12200	335000

"Specific solvation" arises from HB interactions and broadly accounts for ionic reactivity trends

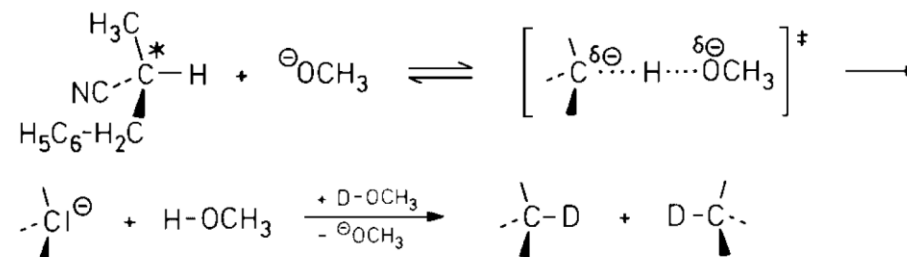
Solvent-ion HB dictates relative reactivity



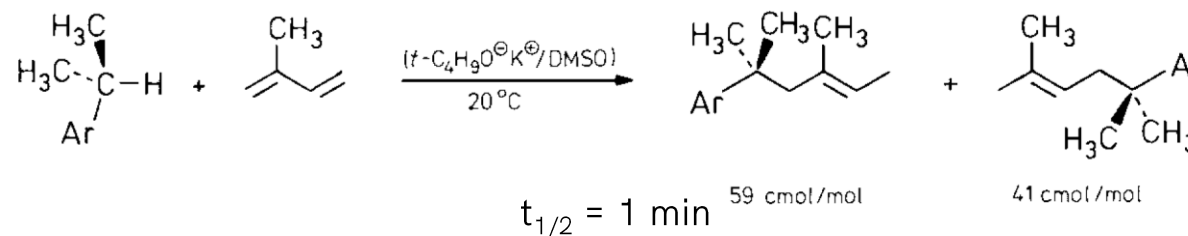
Cl⁻ is inherently most nucleophilic!

in HB solvents, the smaller anion (Cl⁻) is more strongly solvated, diminishing its inherent nucleophilicity

Polar aprotic solvents enhance reactivities of anions

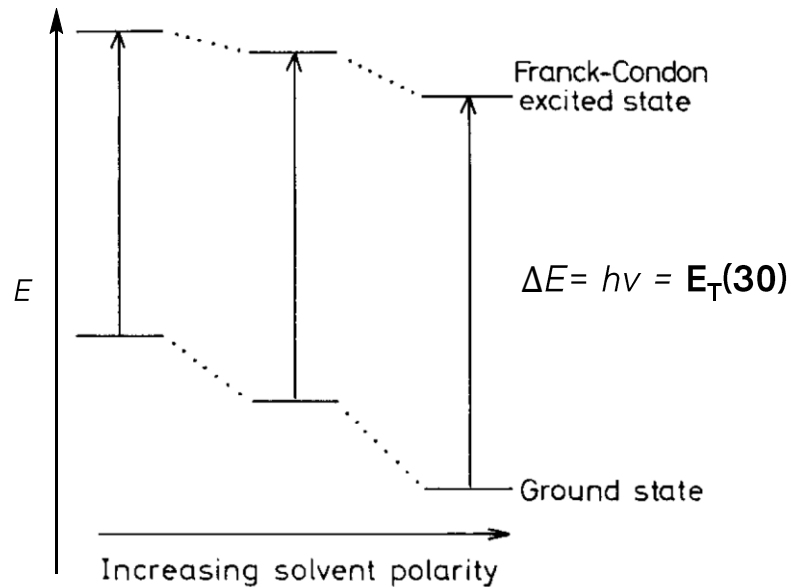
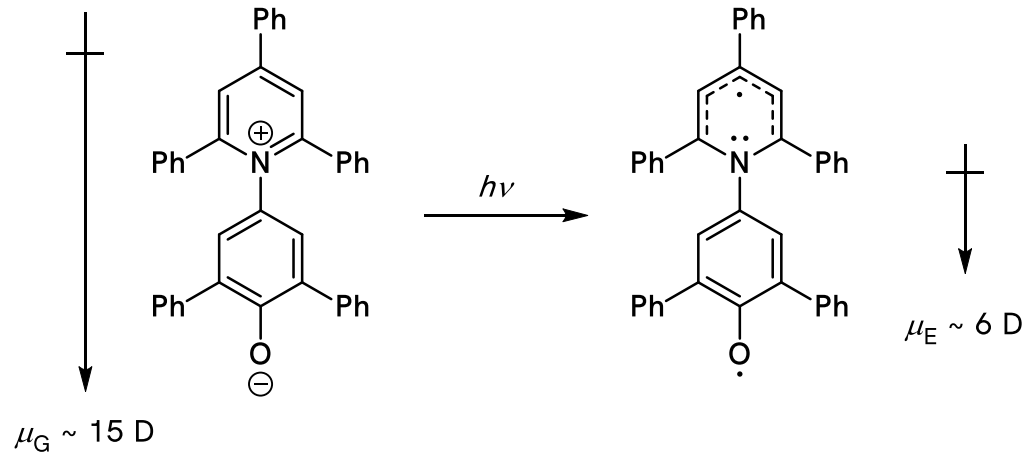


Solvent CH ₃ SOCH ₃ in cg/g	CH ₃ OH/CH ₃ SOCH ₃					
	0	25	50	76	90	98.5
<i>k</i> ₂ ^{rel} (25 °C)	1	32	160	4900	1.3 · 10 ⁵	5.0 · 10 ⁷

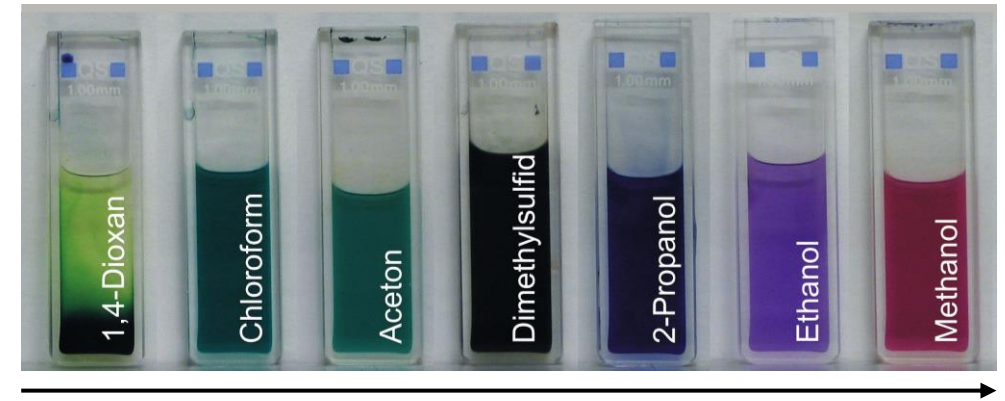


"In solution, the behavior of ions and molecules is dictated mainly by the solvent and less so by their intrinsic properties."

Solvatochromism arises from analogous solvation effects of ES/GS

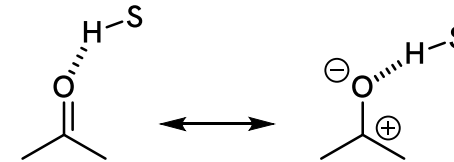


More polar solvent solvates GS better, increasing ΔE



increasing solvent polarity
 blueshift in absorption
 negative solvatochromism

Same principles apply to all types of spectroscopy, e.g. IR:



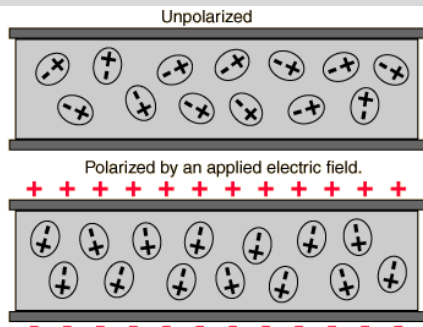
solvent	hexane	DMSO	aniline	water
$\bar{\nu}_{C=O} \text{ (cm}^{-1}\text{)}$	1721	1709	1703	1698

increasing solvation of O^- (ES)
 smaller ΔE
 positive solvatochromism

A good polarity metric should correlate generally with solvent effects on rate/equilibria

Relative permittivity/Dielectric constant ϵ_r

Polarizability of material in an applied electric field



convenient, but does not correlate with experiment

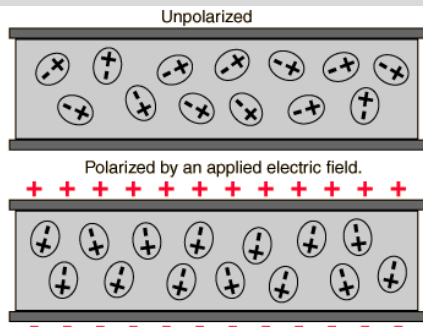
Are the following helpful?

DCM ($\epsilon_r = 8.93$) > THF ($\epsilon_r = 7.58$) > EtOAc ($\epsilon_r = 6.02$)

MeCN ($\epsilon_r = 37.5$) >> AcOH ($\epsilon_r = 6.26$)

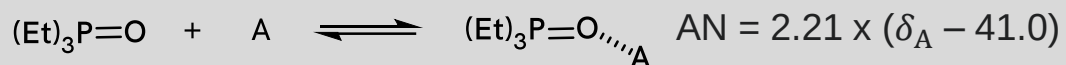
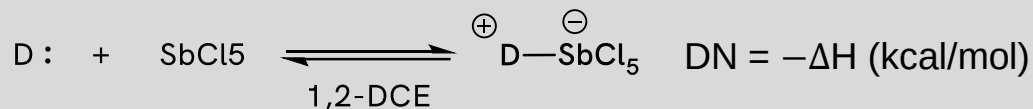
A good polarity metric should correlate generally with solvent effects on rate/equilibria

Relative permittivity/Dielectric constant ϵ_r
Polarizability of material in an applied electric field



convenient, but does not correlate with experiment

Acceptor/Donor Number

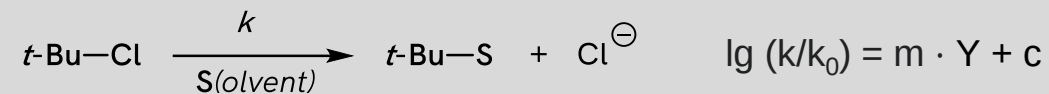


convenient to measure

correlates with E_{HOMO}

measures only Lewis acid/basicity of single solvent molecule

"Ionizing power" Y

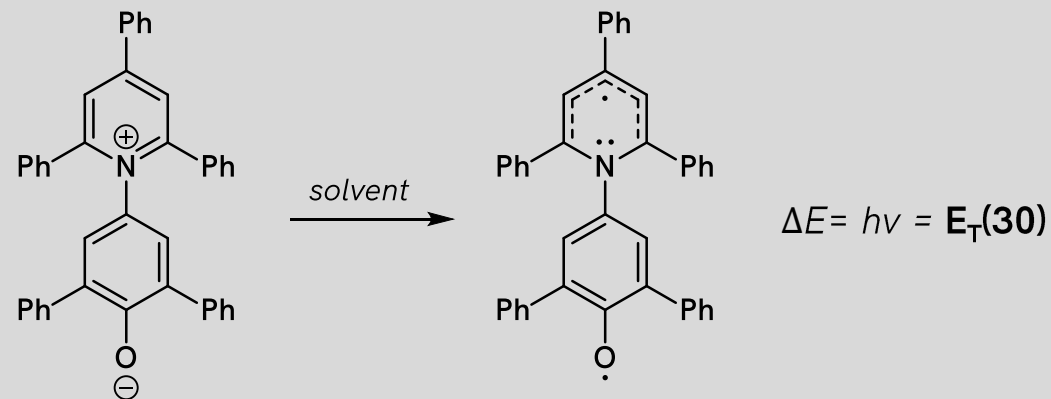


good correlations with S_N1 solvolysis rates

fails for S_N2 (solvent Nucleophilicity matters)

Y is leaving-group-dependent, i.e. many scales required

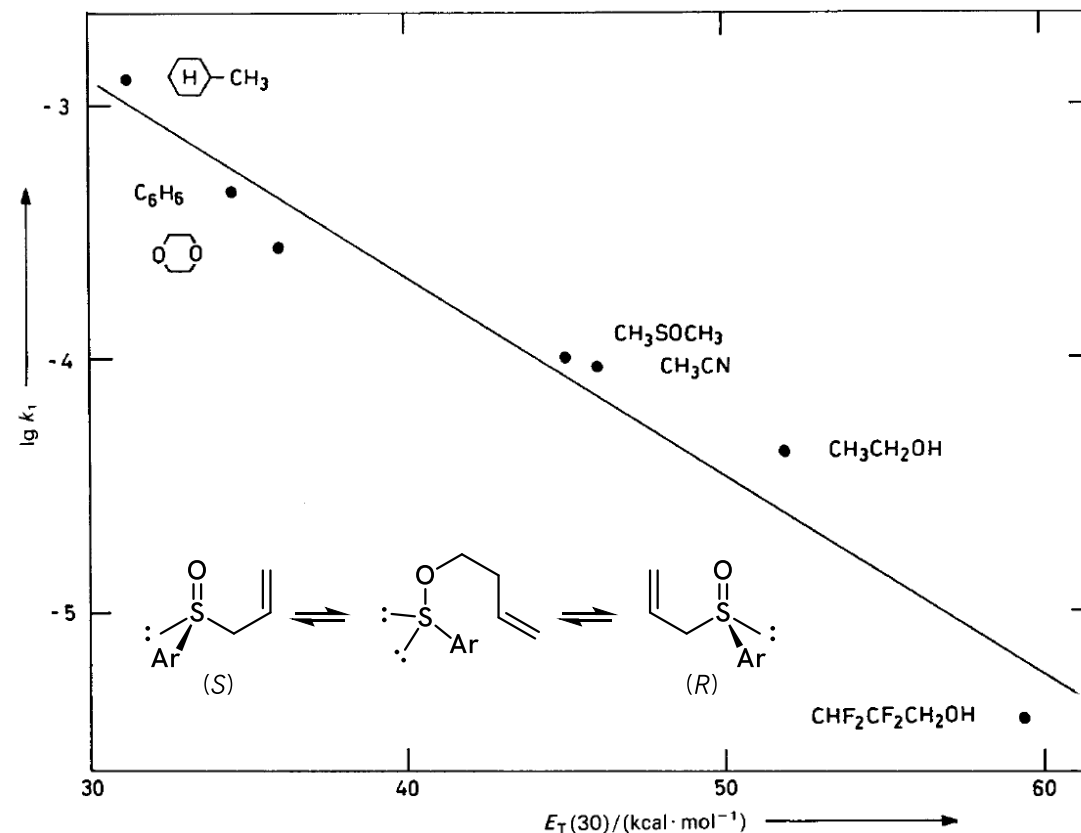
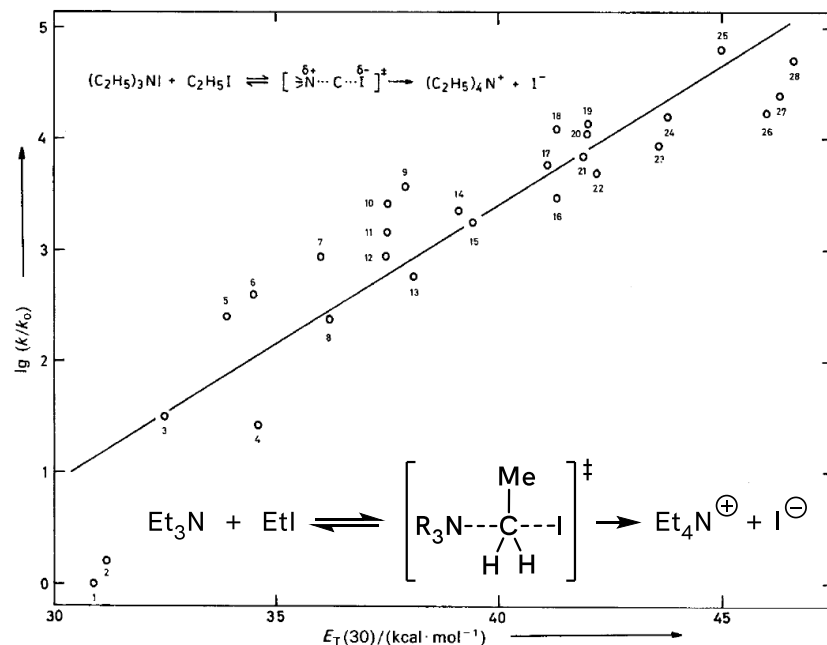
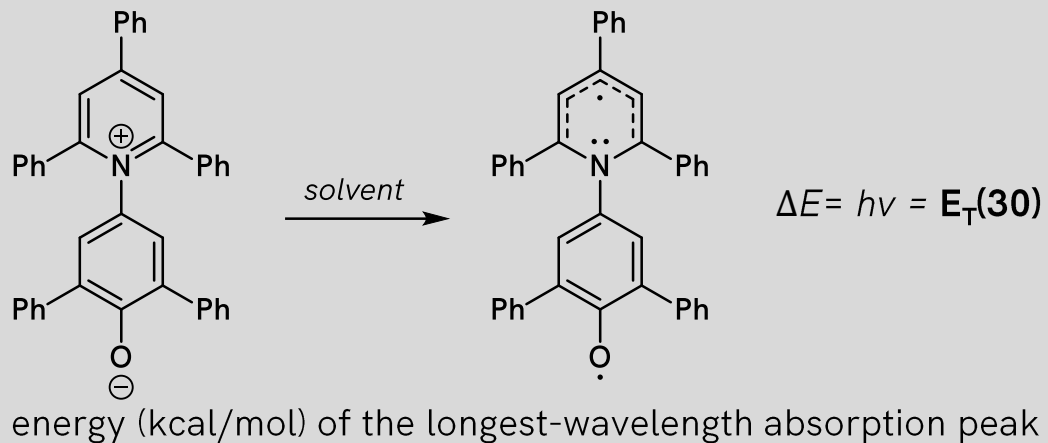
Solvatochromic parameter $E_T(30)$



energy (kcal/mol) of the longest-wavelength absorption peak

A good polarity metric should correlate generally with solvent effects on rate/equilibria

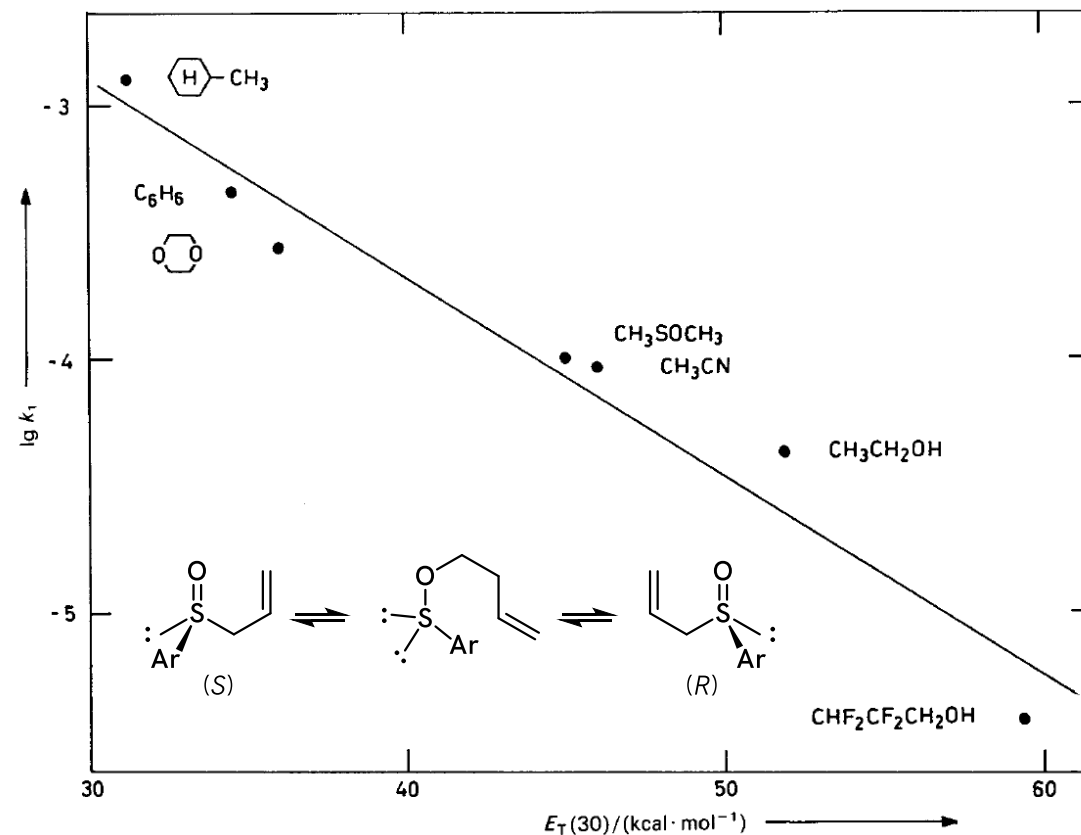
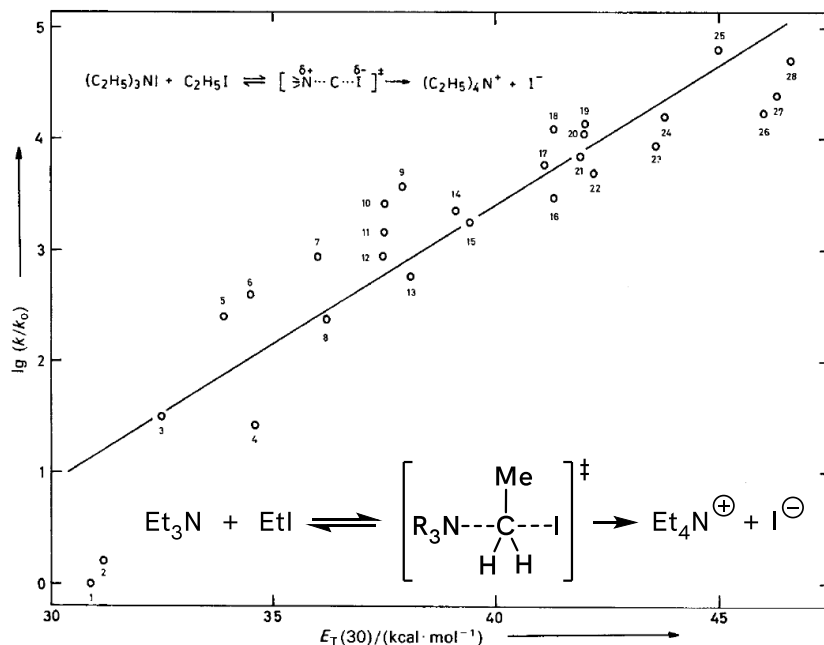
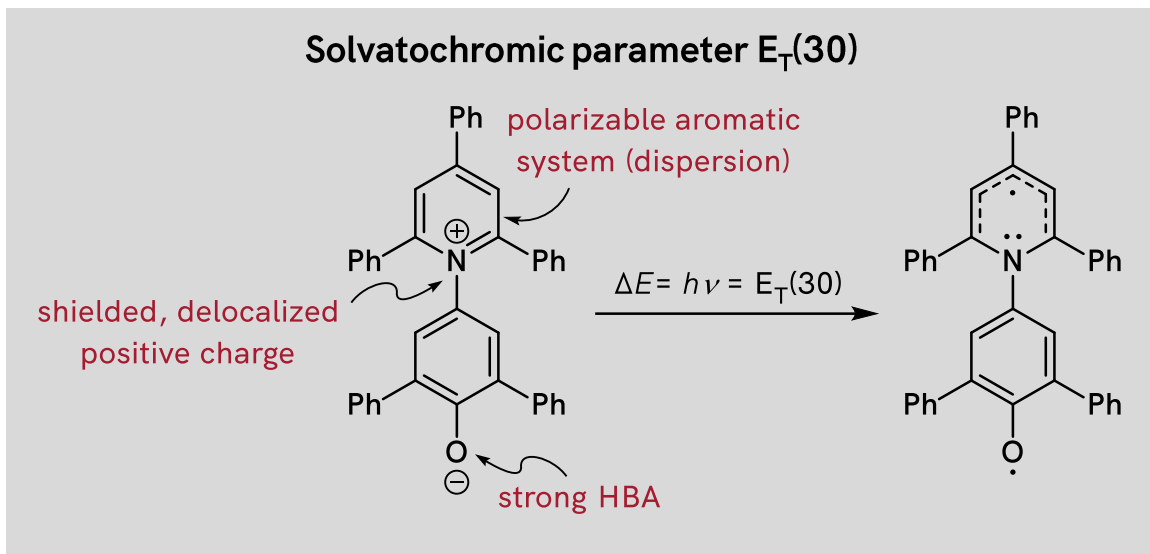
Solvatochromic parameter $E_T(30)$



good correlations with various rate/equilibria

extensively characterized (> 300 solvents, as well as solvent mixtures, electrolytes, micellar solutions, etc.)

A good polarity metric should correlate generally with solvent effects on rate/equilibria



good correlations with various rate/equilibria

extensively characterized (> 300 solvents, as well as solvent mixtures, electrolytes, micellar solutions, etc.)

can contributions from various interactions be separated?

A multivariate LFER-based scale can deconvolute specific solvation properties

Kamlet-Abboud-Taft (KAT) parameters

$$A = A_0 + s(\pi^* + d \cdot \delta) + a \cdot \alpha + b \cdot \beta$$

π^* dipolarity (cyclohexane 0.00, DMSO 1.00) δ polarizability correction (1.0 aromatic, 0.5 polychlorinated, 0.0 rest)

α HBD ability (cyclohexane 0.0, MeOH 1.0) β HBA ability (cyclohexane 0.0, HMPA 1.0)

parameters obtained by fitting across multiple reference reactions

s, d, a, b susceptibilities (analogous to ρ)

The sensitivity of an arbitrary observable to each solvation property can be obtained by multiple regression, e.g.

$$E_T(30) = 11.5 \cdot \pi^* + 15.2 \cdot \alpha + 31.2$$

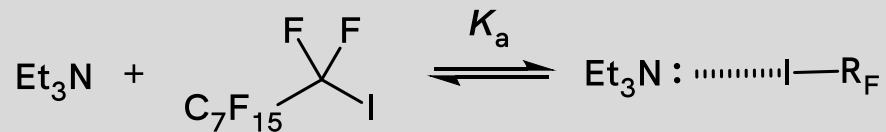
$E_T(30)$ is driven by HBD and polarizability, but has no contribution from HBA

Good to excellent linear correlations obtained for a variety of reactions

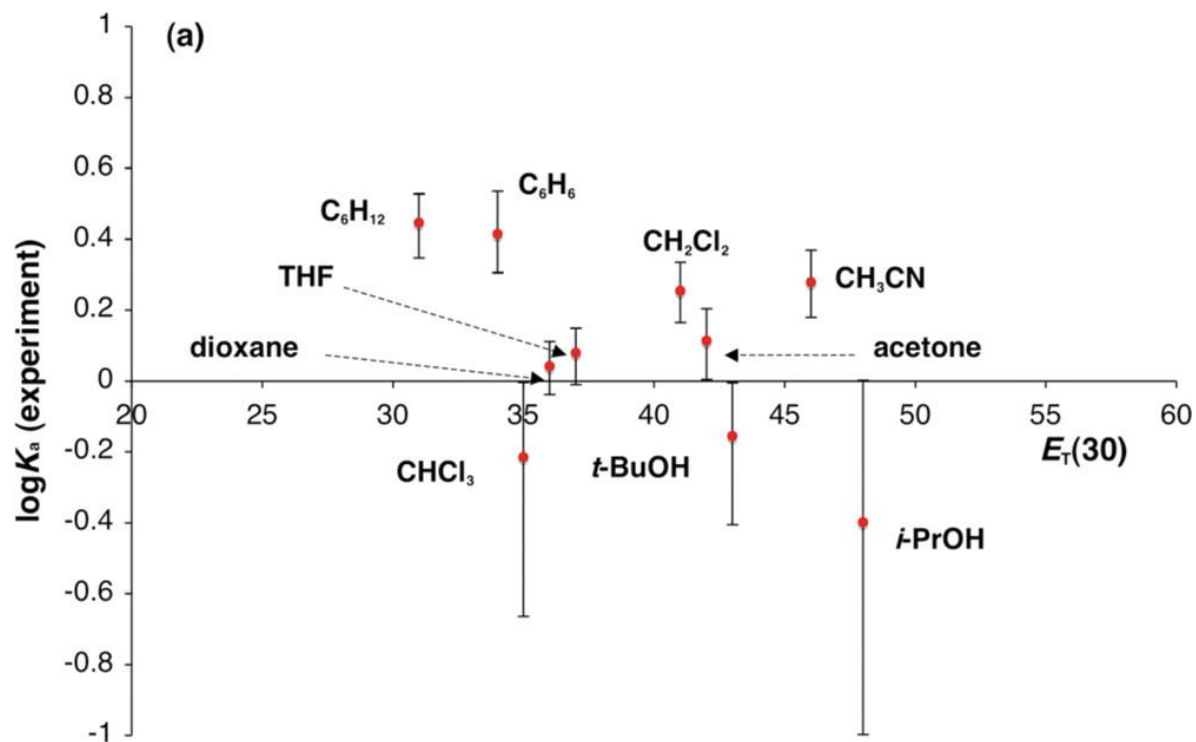
However, so many points are needed for a good correlation that extrapolation is usually impossible

A multivariate LFER-based scale can deconvolute specific solvation properties

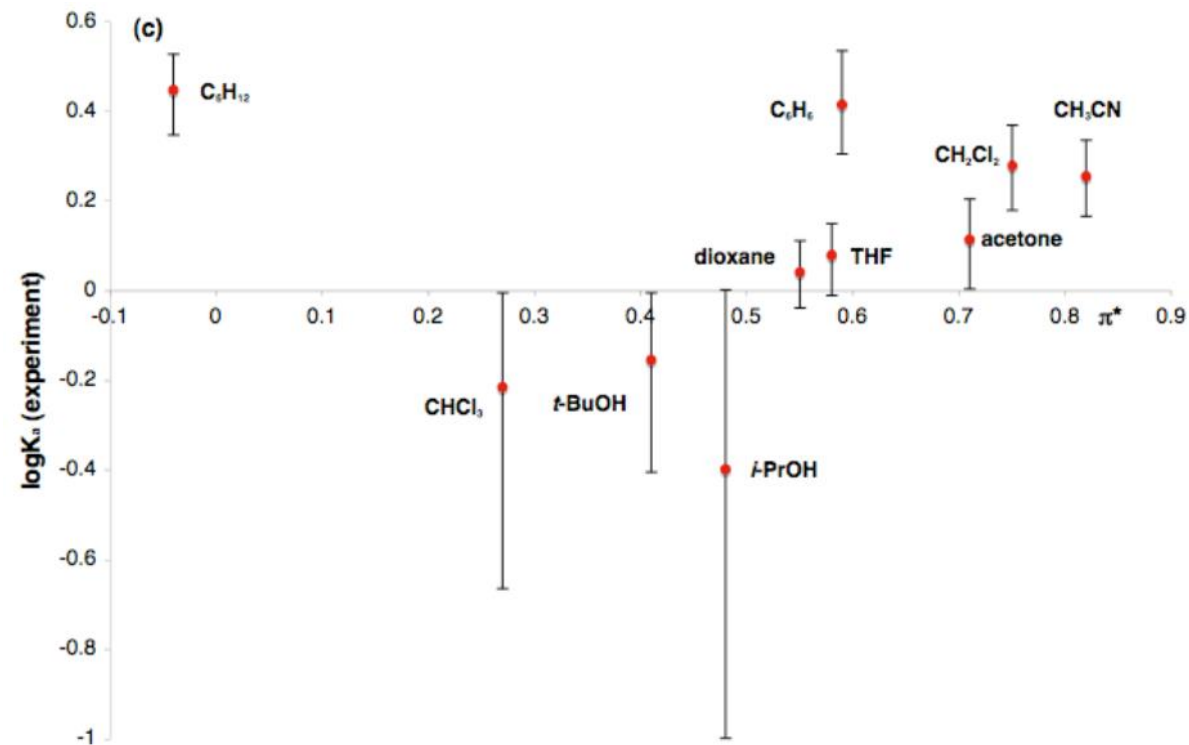
Association of halogen-bonded complex



Plotting against $E_\text{T}(30)$ is not instructive



But plotting against dipolarity reveals a cluster of outliers!



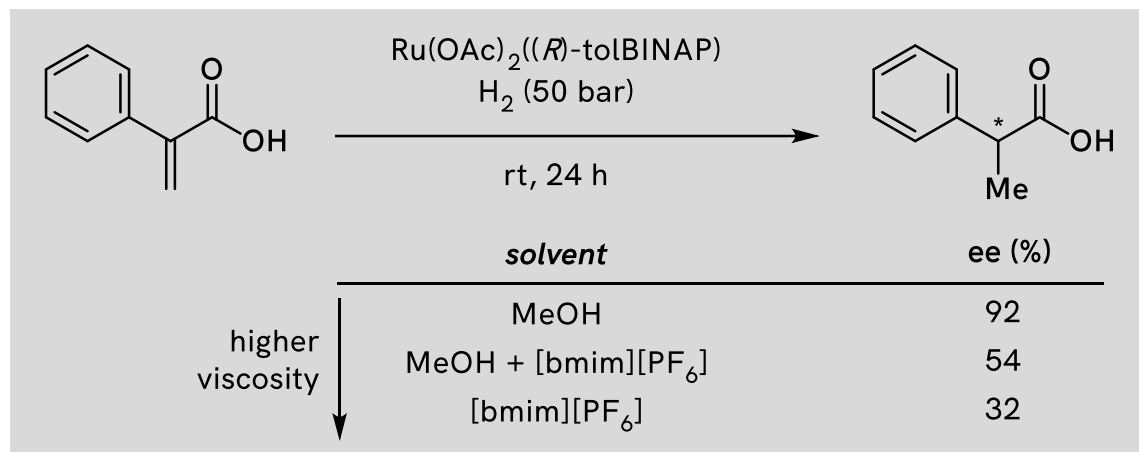
Halogen bonding is relatively insensitive to solvent dipolarity, but HBD solvents are competitive inhibitors

Solvent effects on catalysis

Solvents can affect catalytic reaction outcomes in myriad ways

"Trivial"

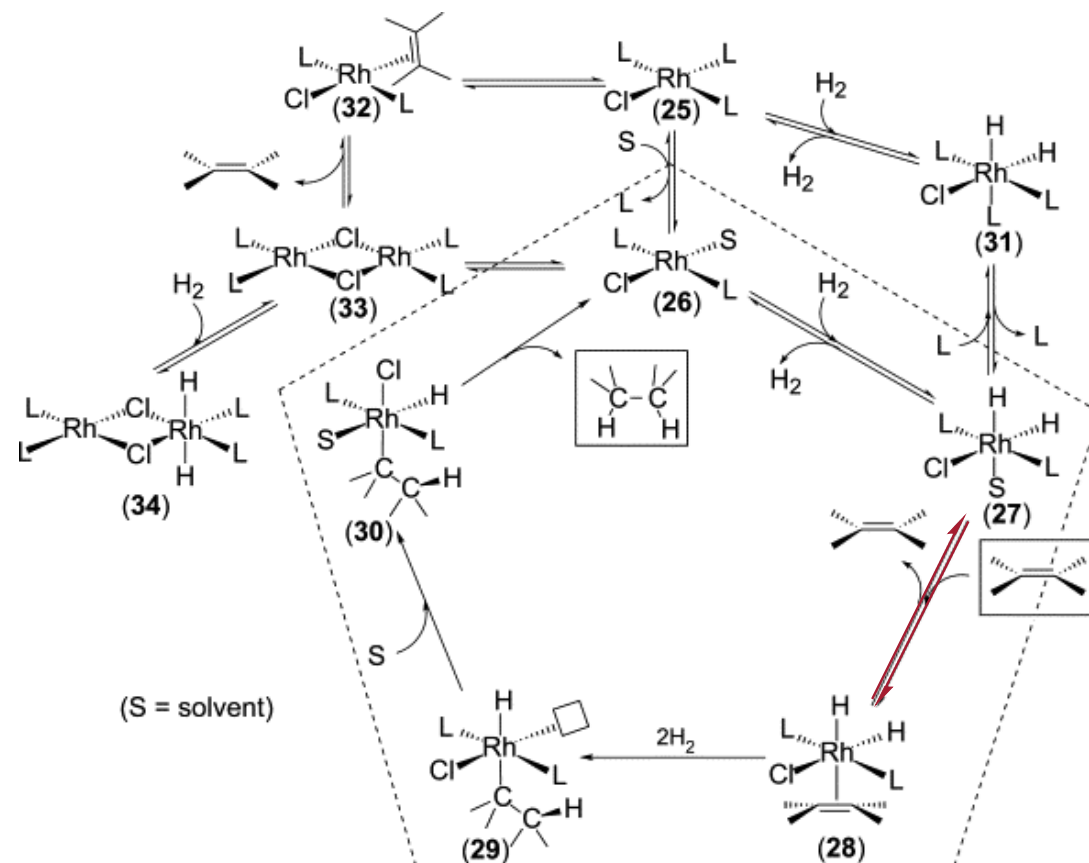
1. Solubility
2. Mass transfer/viscosity



"Nontrivial"

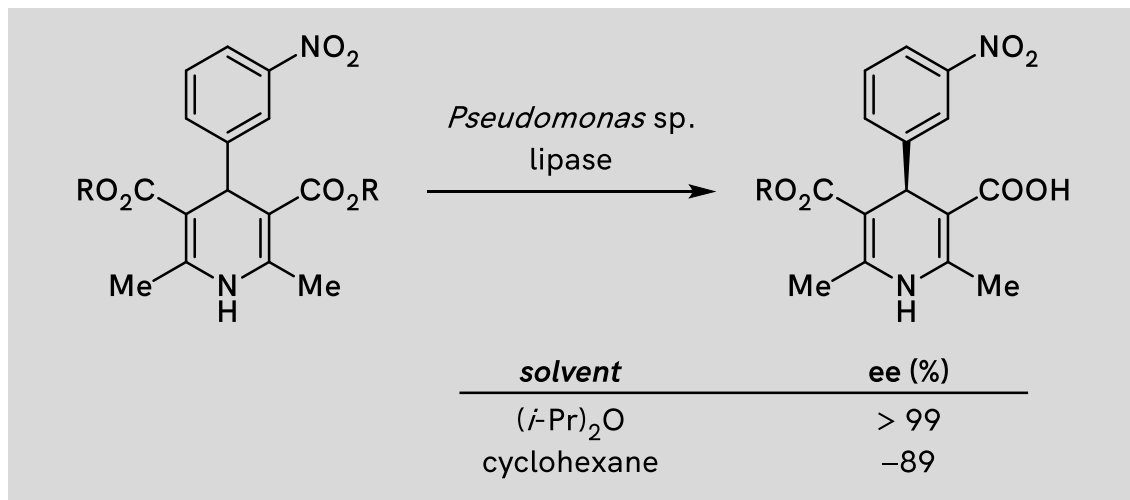
1. Solvent participates in key step/off-cycle pathways
2. Differential solubility of selectivity-determining structures
3. Changes in catalyst conformation/flexibility
4. Non-equilibrium dynamic effects

e.g. Displacement of solvent from metal is rate-limiting



Selectivity in enzyme catalysis is known to depend on solvent

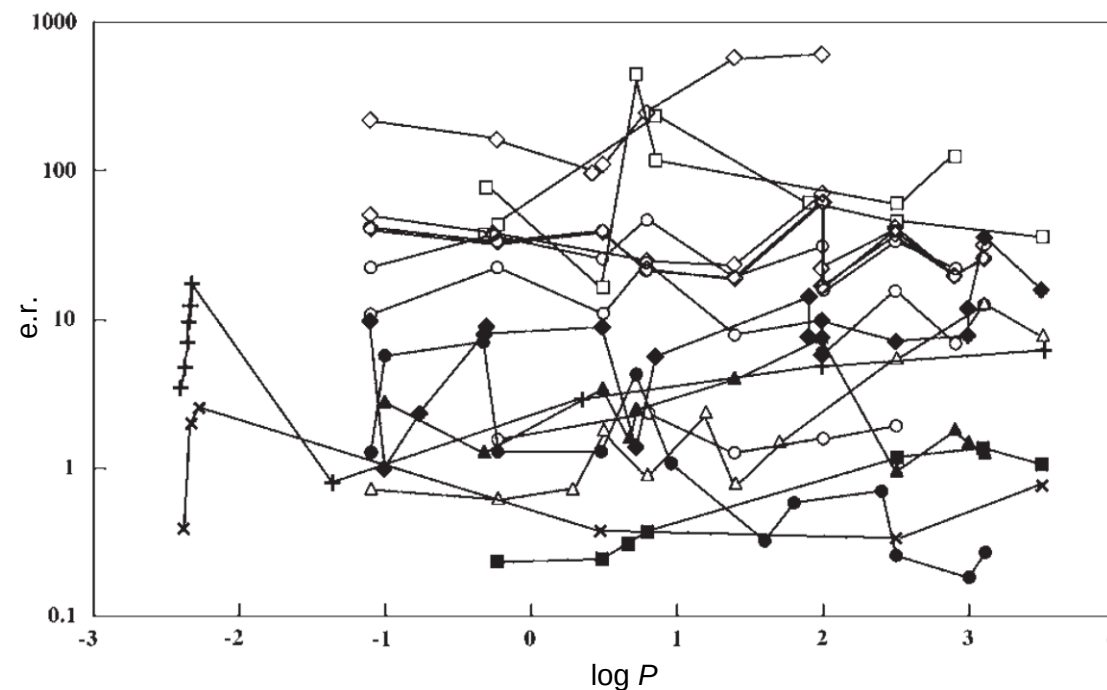
Solvent effects can lead to dramatic enantiodivergence:



Substrate specificity, chemoselectivity, and regioselectivity can all be altered with solvent choice

Hirose & Achiwa *TL* **1992**, 33, 7157; Klibakov *Nature* **2001**, 409, 241

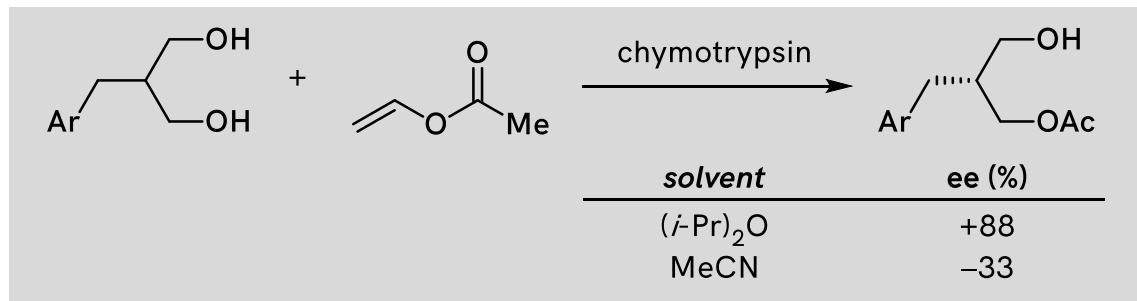
Variations in selectivity do not correlate with "polarity":



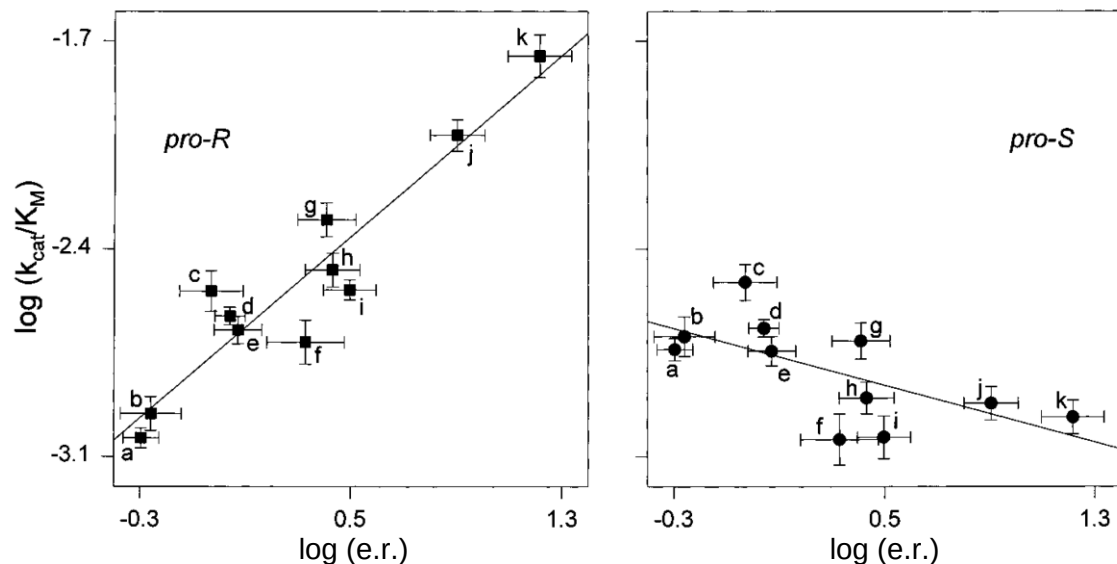
Jongejan *Organic Synthesis with Enzymes in Non-Aqueous Media* **2008**

Proposed explanations include change in enzyme structure/conformational flexibility, competitive solvent binding, and differential solvation of TSs.

Solvent effect on desymmetrizations is proposed to arise from differential solvation of diast TSs



As solvent is varied, major pathway is accelerated, minor pathway is decelerated

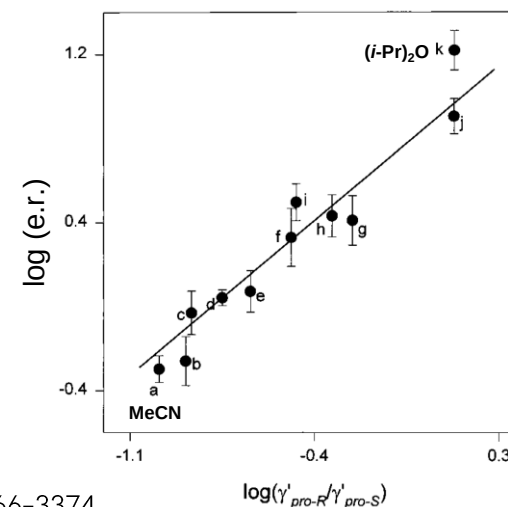
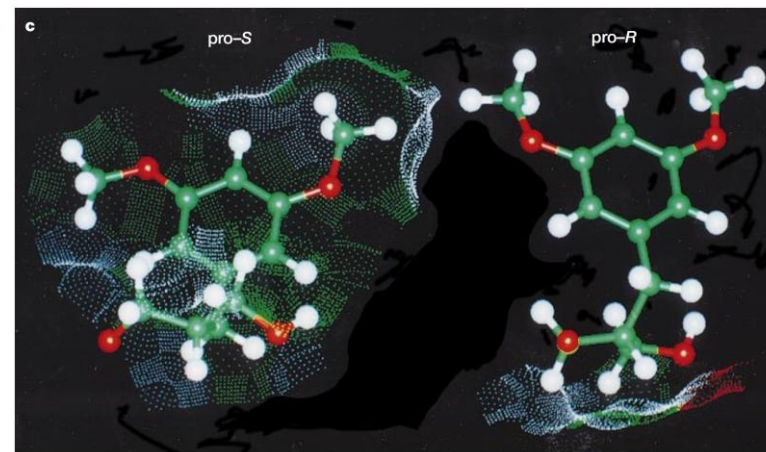


Solvents that lead to higher selectivity also lead to higher rate!

No correlation between solvent polarity and ee/rate.

Interpretation

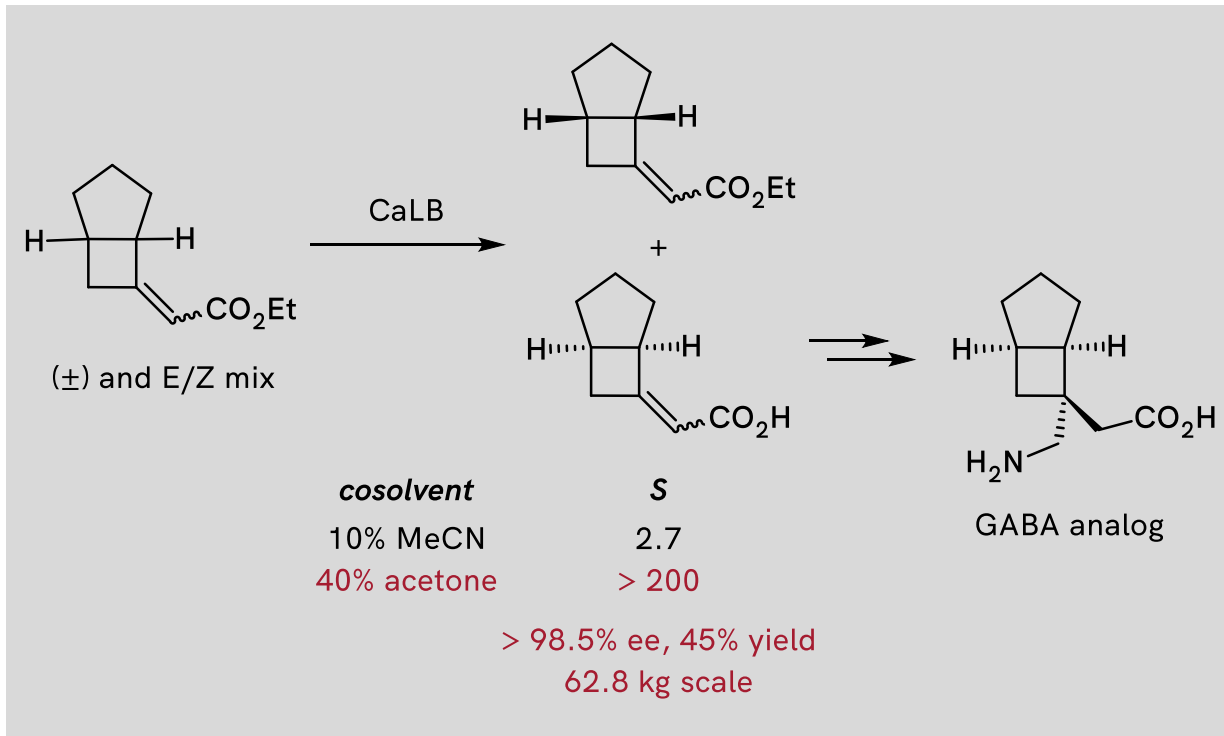
In *pro-R* and *pro-S* TSs, different parts of substrate are buried in active site/solvent-exposed. Selectivity arises from ratio of solvation energies* of solvent-exposed substrate.



* activity coefficient

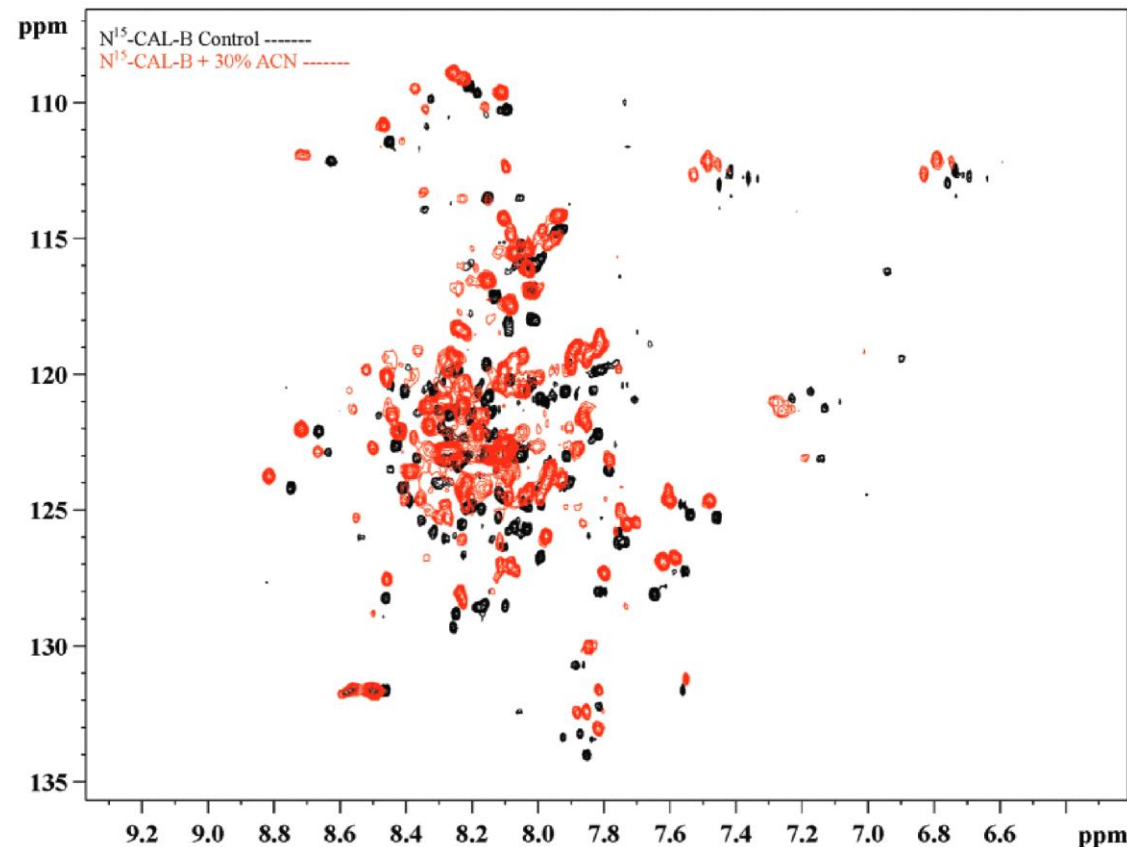
Solvent effect on enantioselectivity is practically enabling

Objective: high *S* factor, low specificity between *E*- and *Z*-substrates



Cosolvent screening transforms non-viable enzyme candidate into a hit

^{15}N TROSY with addition of MeCN shows shifts in amide signals, consistent with a change in enzyme conformation in altered solvent



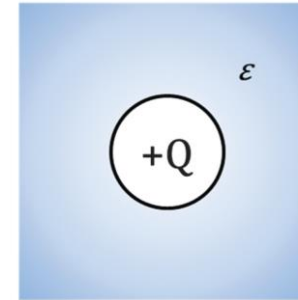
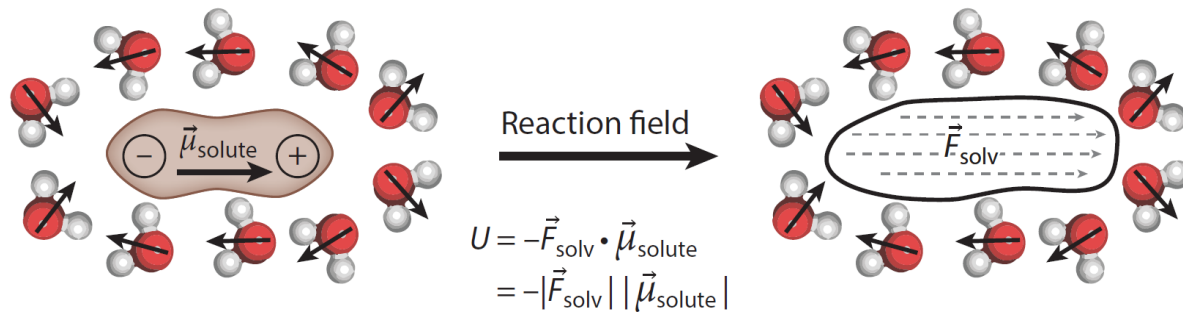
Computational modeling of solvent

Implicit/continuum models

Hybrid (implicit model + explicit solvent) approaches

Explicit solvent

Implicit/continuum models represent solvent as an isotropic field



Solute dipole forces solvent molecules to align
 Aligned solvent molecules create a nonzero field (the “reaction field”)
 Interaction between dipole and field leads to solvation energy U

instead of modeling many configurations of solvent, the continuum treatment serves as a cheap “statistical average”

$$G(\text{solute}) = G_{\text{cav}} + G_{\text{el}} + G_{\text{disp}} + G_{\text{rep}} + G_{\text{tm}}$$

Solvation energy includes cost of orientation and polarization of the solvent, solute dipole moments

$$G_{\text{P}} = -\frac{1}{2} \left(1 - \frac{1}{\epsilon} \right) \sum_{k,k'}^{\text{atoms}} q_k q_{k'} \gamma_{kk'}$$

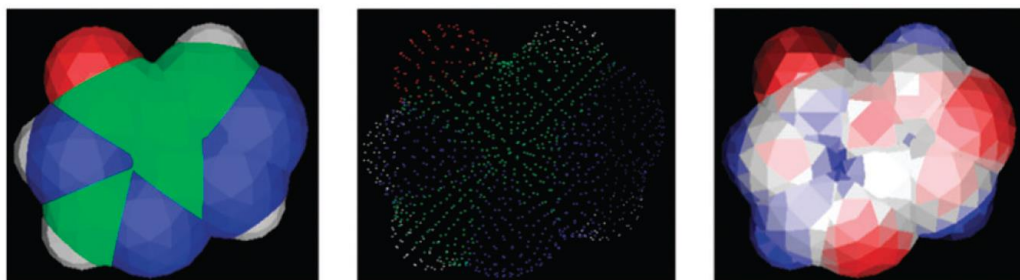
Implicit/continuum models do not have a physical basis

Widely used models, parameterized for common solvents:

1. IEFPCM (*not* CPCM) (SCRF=PCM in Gaussian)
2. SMD (SCRF=SMD in Gaussian)
3. COSMO/COSMO-RS (Q-CHEM)

define solvent-accessible surface area (SASA), then

evaluate reaction field, polarization energy at selected points:

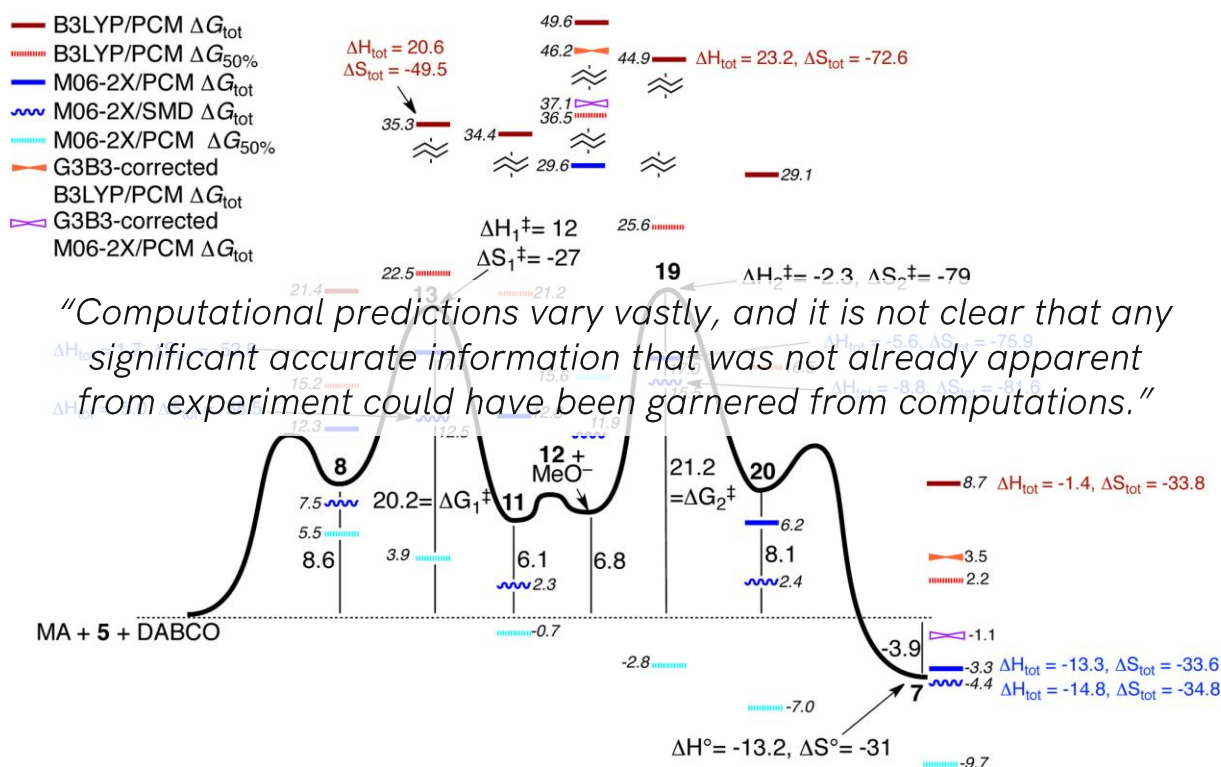
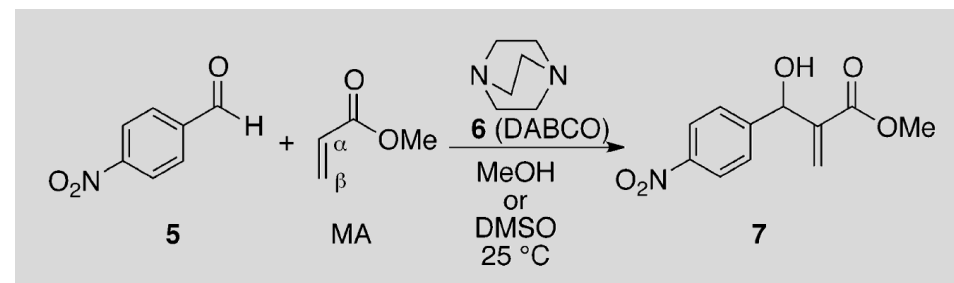


Solvation energy is highly dependent on the shape of the surface!

Parameterization involves arbitrarily scaling atomic radii used to fit (limited) experimental data.

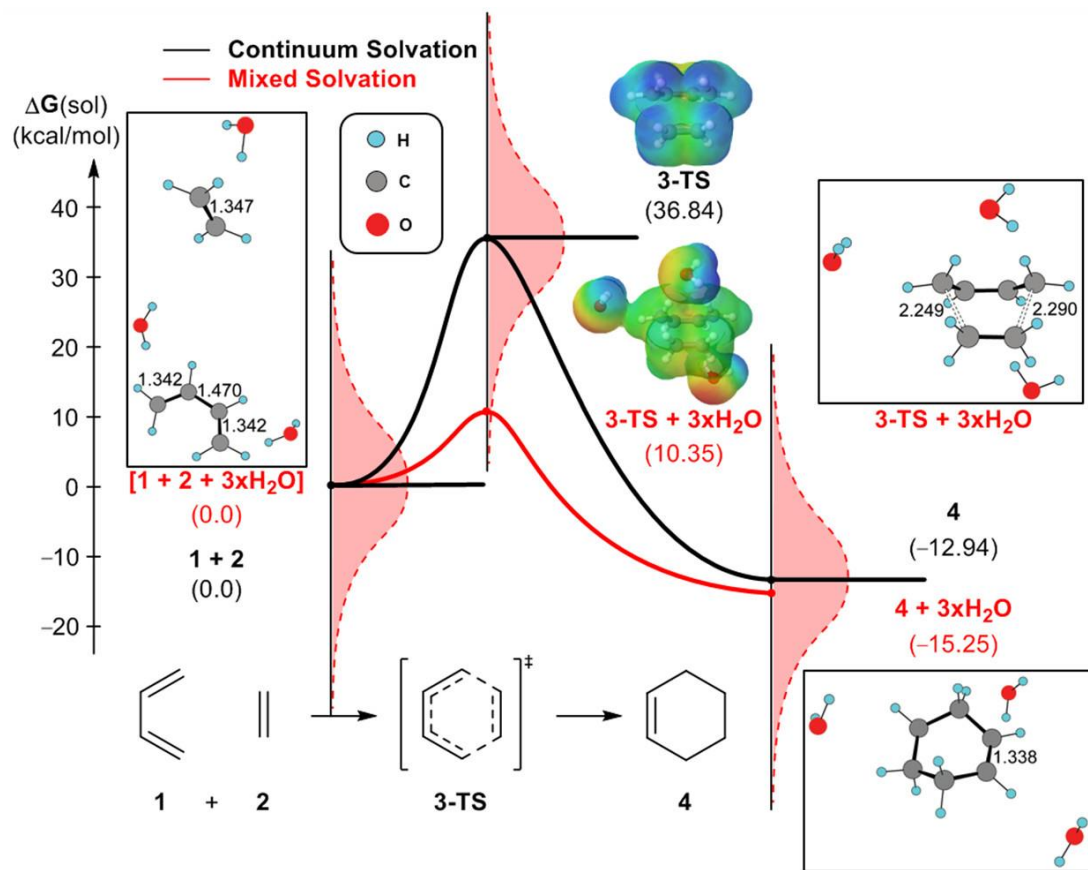
"They are oversimplified fit parameters that reduce the complexity of solvent-solute interactions [...] into a fictitious surface potential, the physical meaning of which derives from being scaled to reproduce experimental solvation free energies."

Implicit solvent models completely fail at predicting relative energies along MBH reaction coordinate

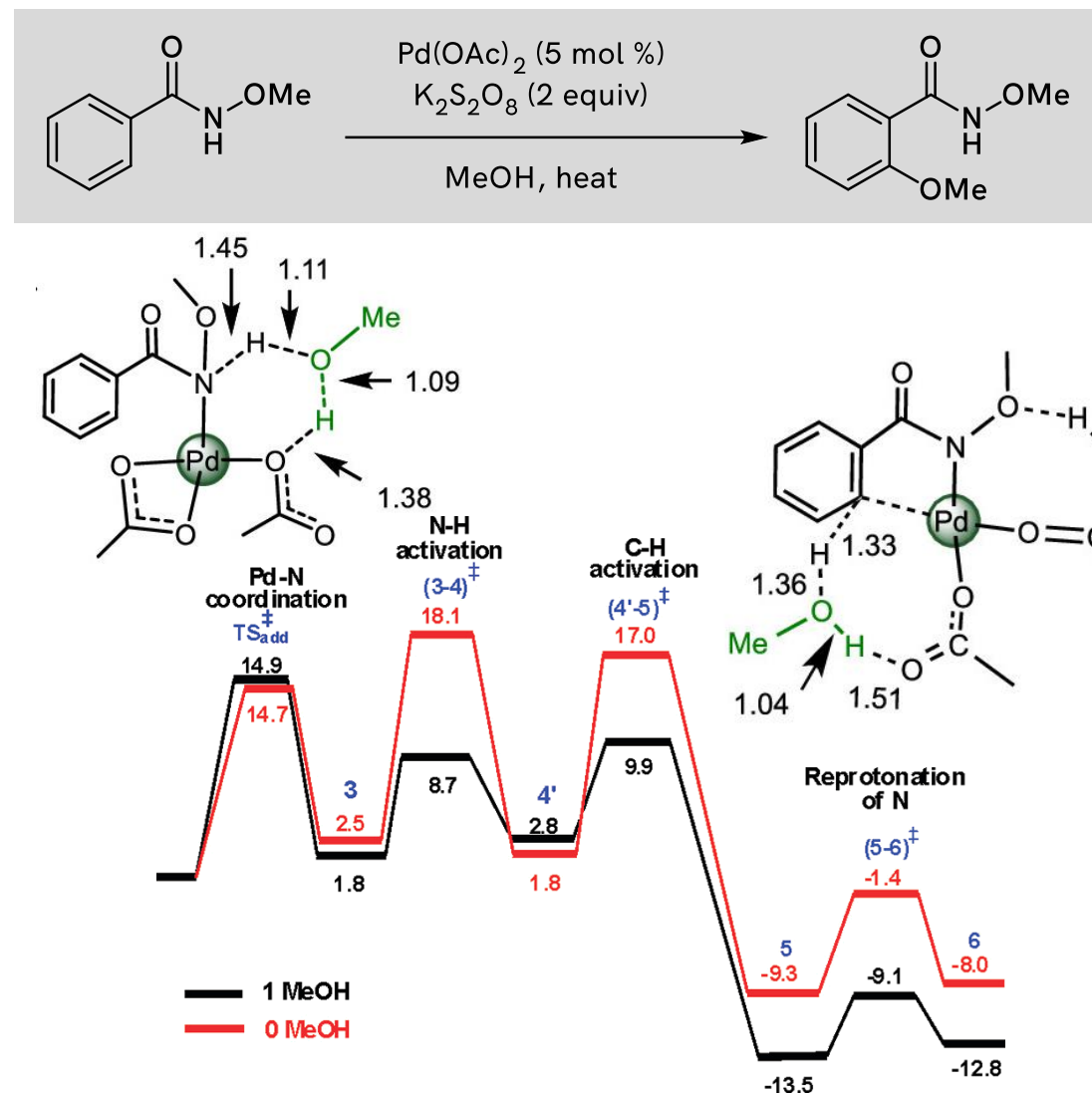


A step up (down): adding a few explicit solvent molecules is not advisable unless they participate

Adding a small number of solvent molecules around reactants to mimic specific solvation interactions sounds reasonable but is not:



Explicit solvent is valid when they participate in the reaction



The explicit solvent approach is costly but physically realistic

Use cases

Ideally: every case

Practically: when subtle solvent effects elude simpler solvation models and/or experimental investigation
and
when computationally tractable

Adding dozens–hundreds of explicit solvent molecules leads to a rugged PES; there are no longer single minima.

Single-saddle-point optimizations are pointless

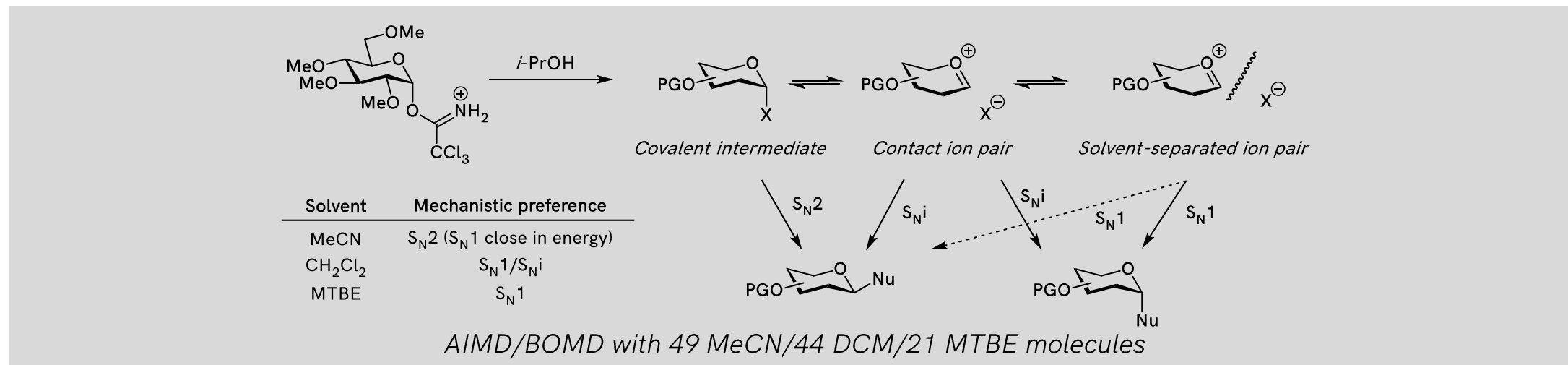
Molecular dynamics simulations become necessary

MD is costly: each frame is a full geometry optimization

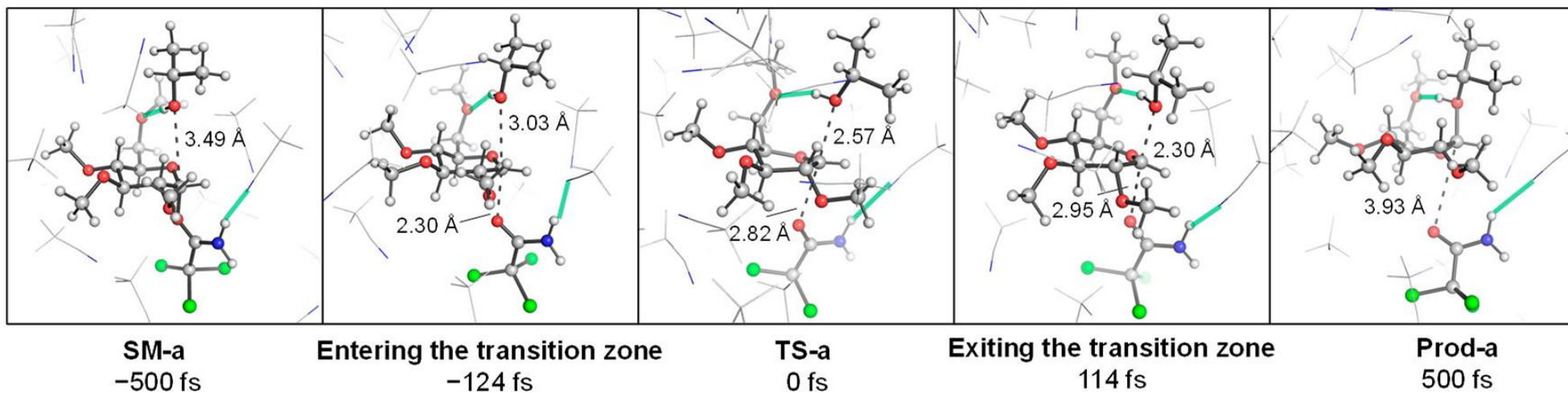
Potential origins of “complex” solvent effects:

1. Solvent noninnocence or multiple solvent participation
2. Dynamic effects
 - When chemical events are faster than solvent reorganization
 - “Entropic intermediates”
 - etc.

AIMD with explicit solvent illuminates the role of solvent in chemical glycosylations

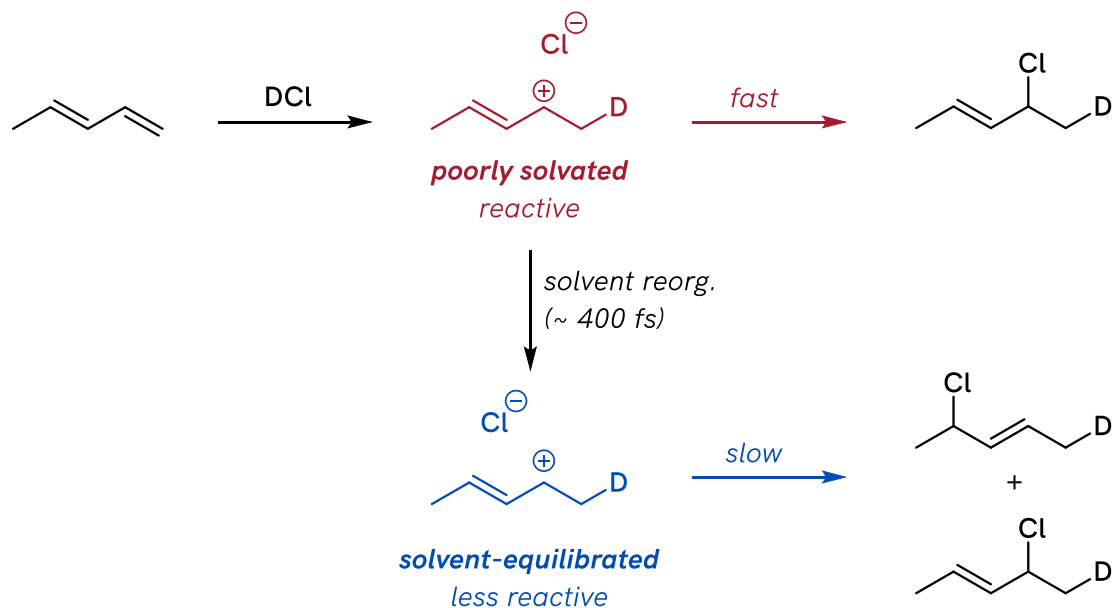
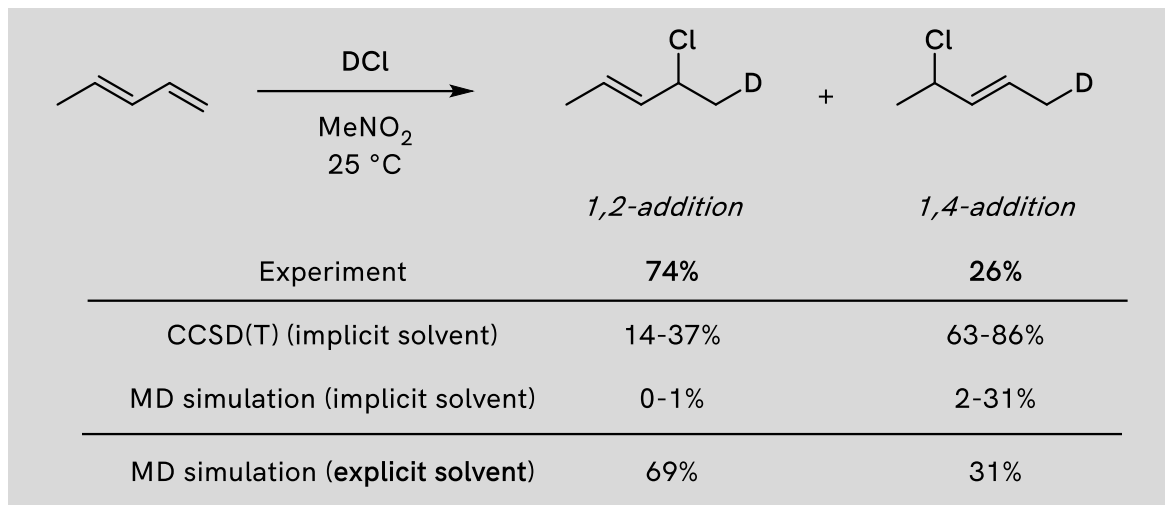


in MeCN:

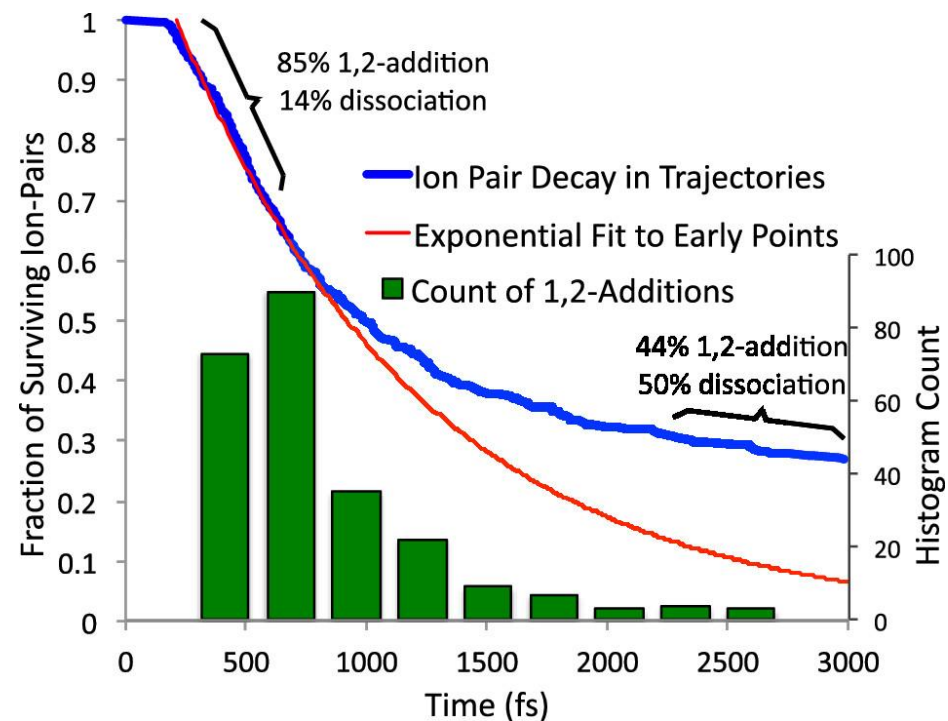


The role of specific solvent molecules in stabilizing various species on the trajectory is otherwise impossible to probe

AIMD with explicit solvent reconciles experimental observations with “frictional” solvent effects



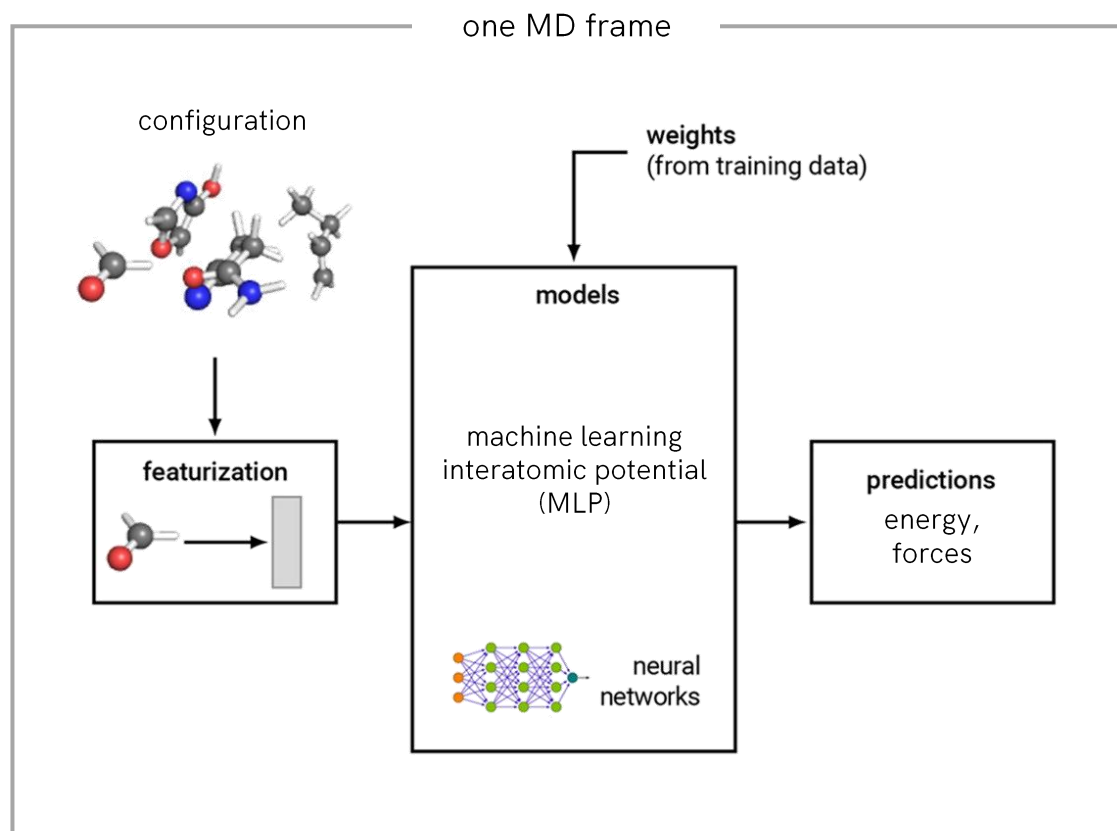
AIMD with multilevel (ONIOM) scheme:
 reactants: M06-2X or MP2
 solvent molecules: semiempirical



before solvent reorg, reactive ion pair collapses into 1,2 adduct
 upon reorg, ion pair prefers dissociating (reverting) over 1,4 adduct

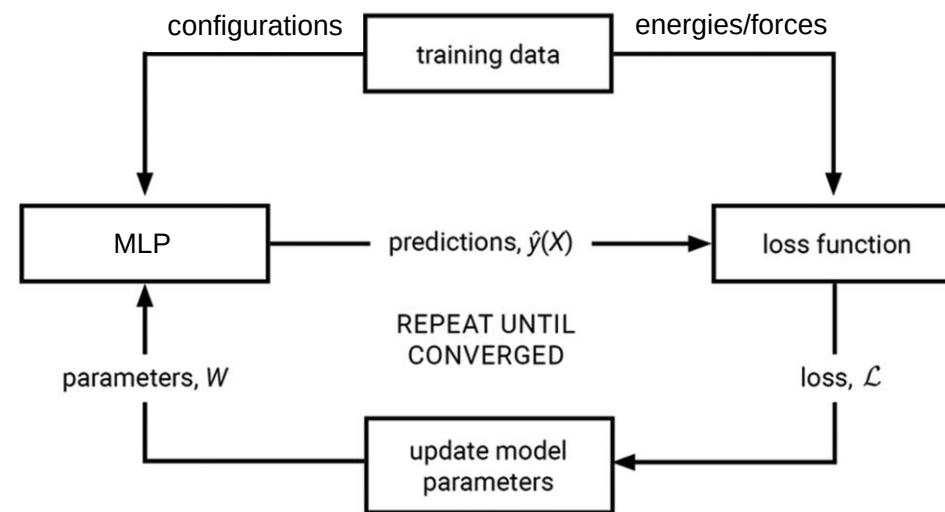
Explicit solvent calculations are expensive, but machine learning potentials can help

Machine learning potentials can predict energy/forces from atomic configurations

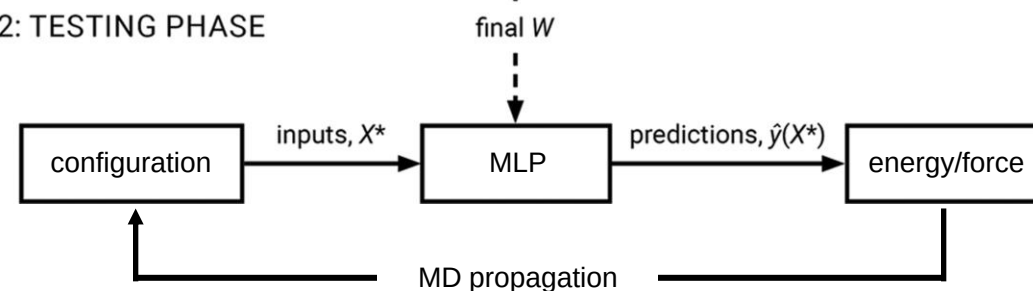


A system-specific MLP can be trained by generating data with DFT/higher

1: TRAINING PHASE



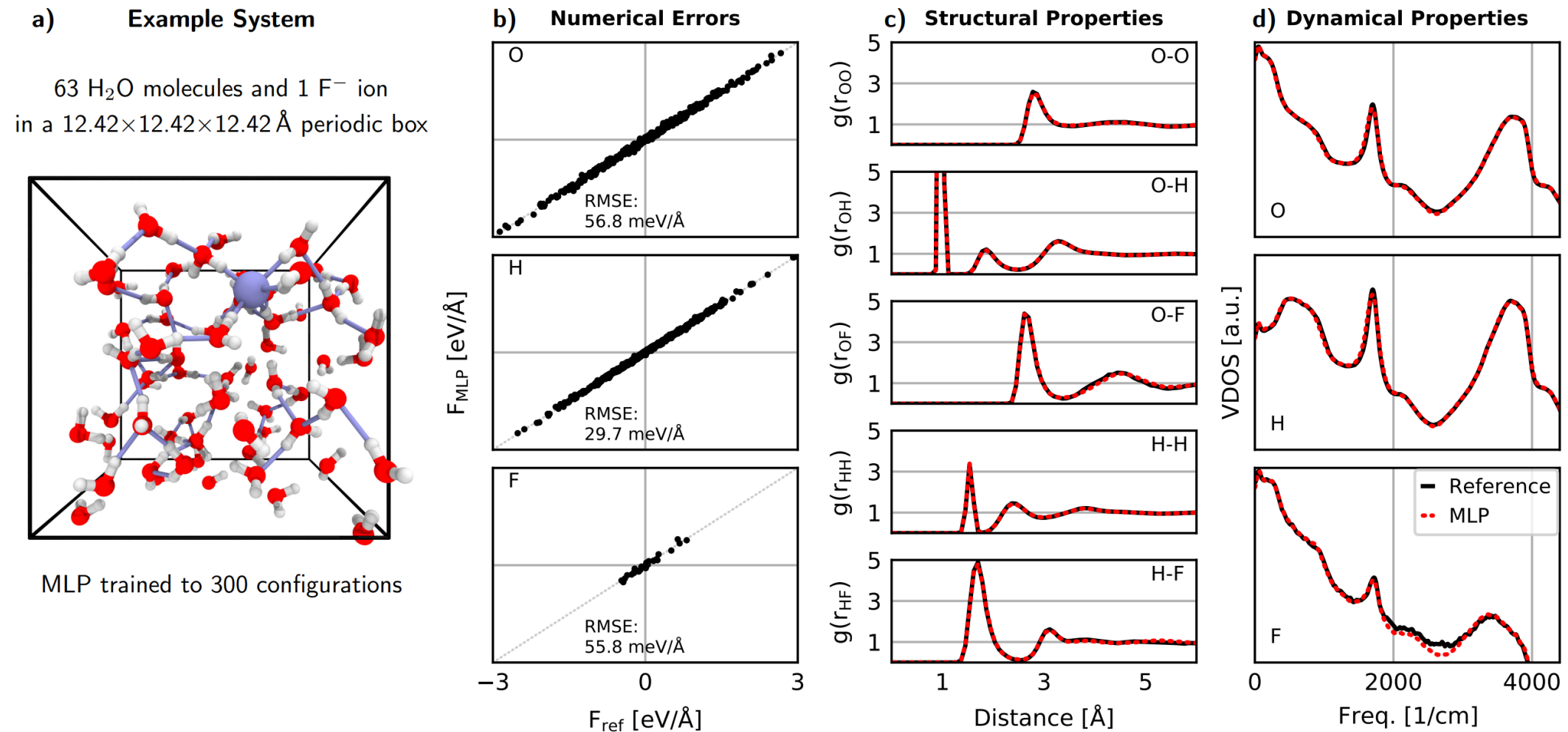
2: TESTING PHASE



But lots of DFT calculations with explicit solvent are needed!

Can we pick **only the most informative configs** for DFT calculation?

Naïve MLP approach is effective in modeling explicit ion solvation

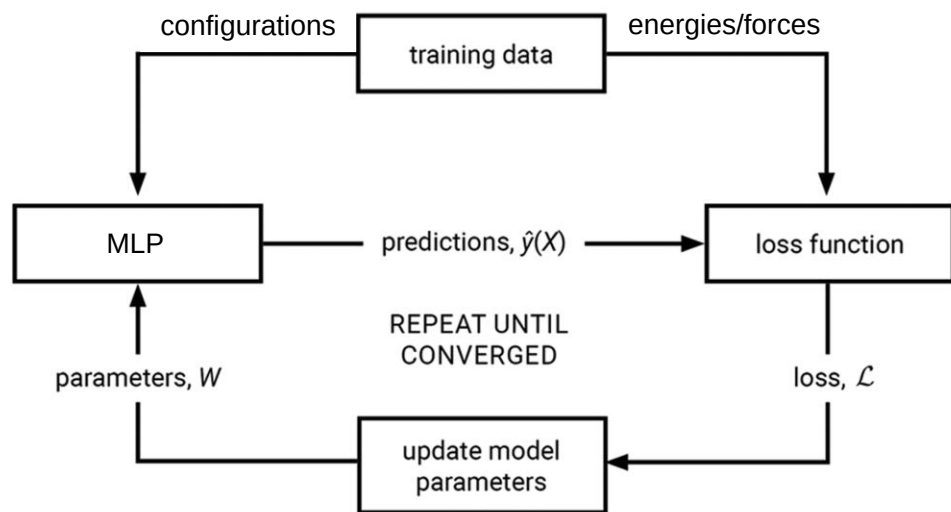


Other demonstrated applications include interfacial behavior, phase changes in bulk materials, nanoconfined materials, etc.

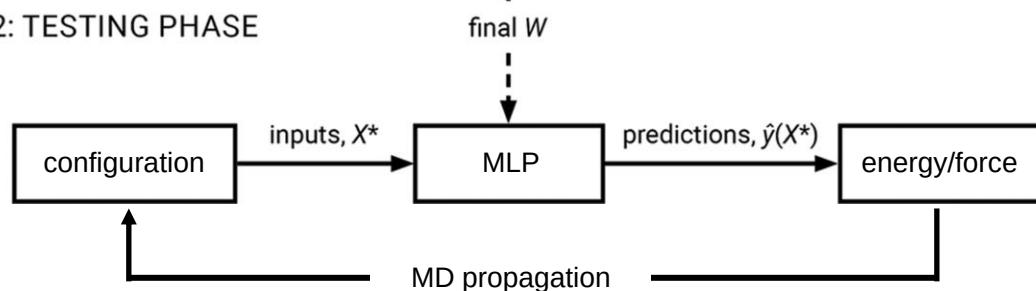
Active learning enables iterative improvements of MLP at much lower cost

Traditional MLP training

1: TRAINING PHASE



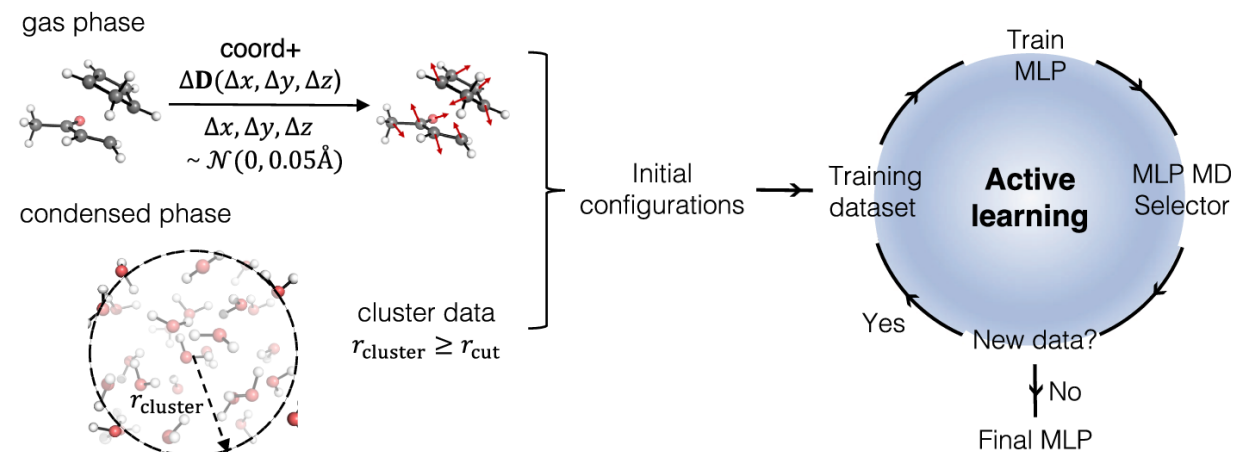
2: TESTING PHASE



many configs required to learn solute-solvent interactions

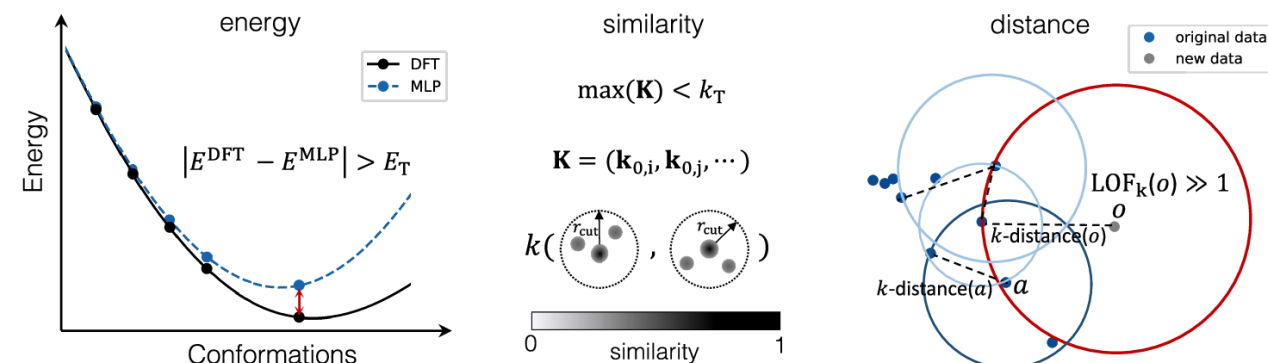
MLP with active learning

a Active learning workflow

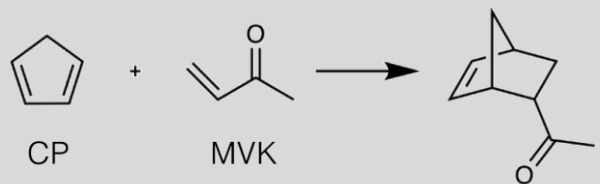


Automated, judicious selection of a small subset of configs for learning

b Selectors



Explicit solvent MD with MLP + AL provides physical insight into solvent effects

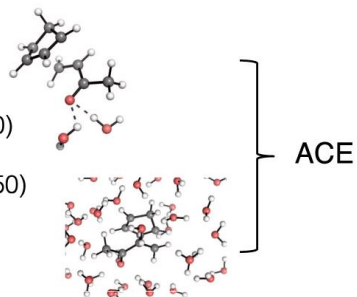


$$k_{\text{water}}/k_{\text{MeOH}} = 58$$

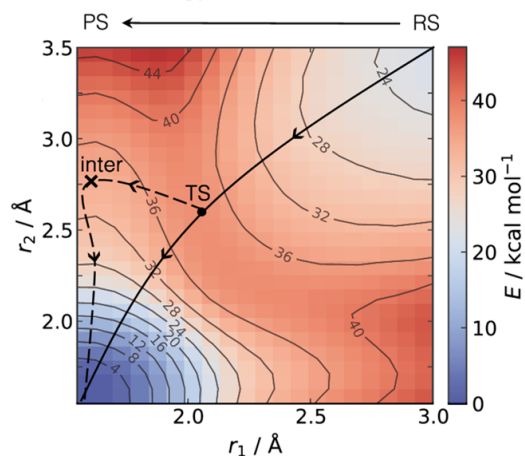
proposal: water (and not MeOH) preferentially stabilizes TS

Sub-training sets

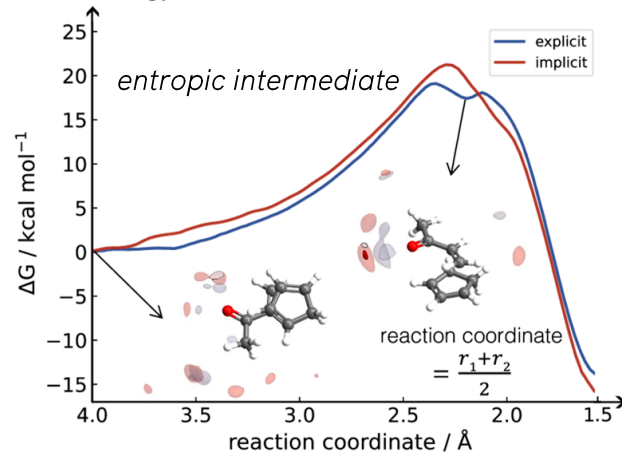
- 1) substrate in gas phase (#150)
- 2) substrate + 2 water molecules (#150)
- 3) substrate + 33 water molecules (#250)
- 4) 27 water molecules (#50)



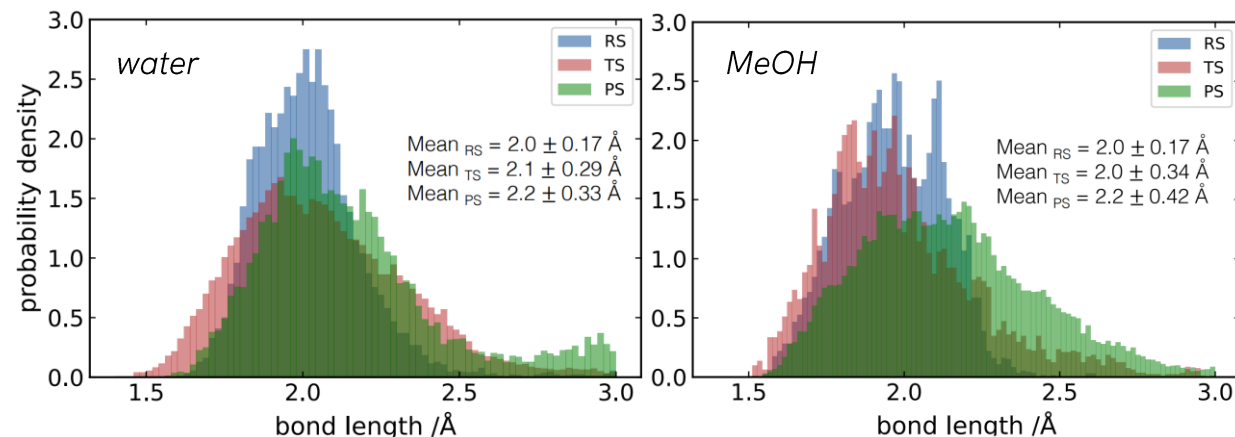
a Potential energy surface



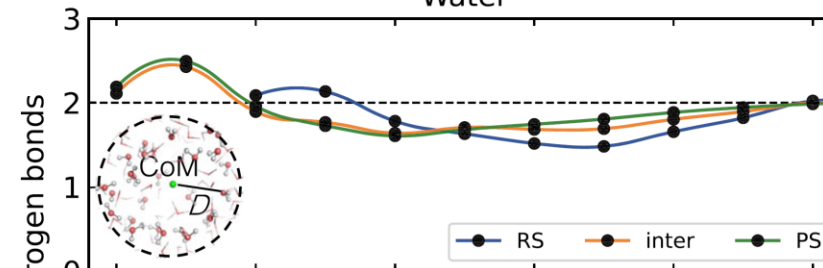
b Free energy surface



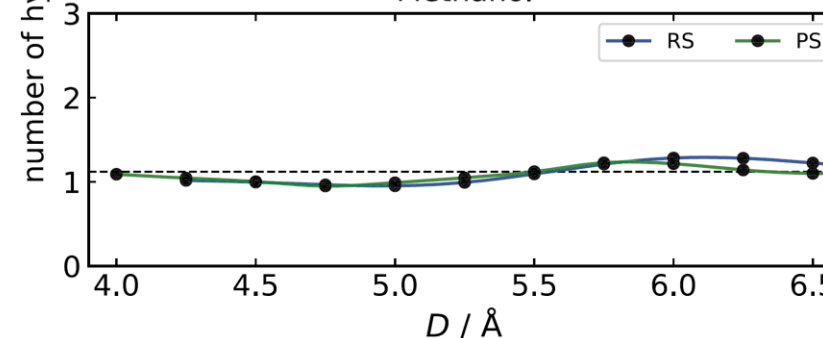
number and strength of HBs stay constant throughout trajectory



Water



Methanol



water is more organized around reacting species

proposal: hydrophobic effect drives reaction in water